TS5000 Robot Controller

Alarm Manual

Rev.6

March 1, 2024

IMPORTANT

(1) Ensure that this manual is provided to the actual final user of the industrial robot.

(2) Be sure to read this manual before operating the industrial robot.

(3) After reading this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

The controller supported by this instruction manual is the TS5000.

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Introduction

Thank you for purchasing this SHIBAURA MACHINE robot system.
This manual describes the alarms, their causes, and corrective actions for the TS5000 controller.

This manual consists of the following sections.

Part 1: Overview
This part gives an overview of the alarm.

Part 2: Details
Chapter 1: Possible Power-on Faults
This chapter describes faults that are not displayed as alarms at power on.

Chapter 2: Level 8 Alarms
This chapter describes level 8 alarms, their causes, and corrective actions.

Chapter 3: Level 4 Alarms
This chapter describes level 4 alarms, their causes, and corrective actions.

Chapter 4: Level 2 Alarms
This chapter describes level 2 alarms, their causes, and corrective actions.

Chapter 5: Level 1 Alarms
This chapter describes level 1 alarms, their causes, and corrective actions.

Manufacturer

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If you have purchased this robot outside of Japan, please contact the company that you purchased it from.

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Usage Scope

This system was designed for supporting automation of a wide range of operations, including transportation operations such as workpiece movement and alignment and assembly operations such as part insertion and screw tightening.
We ask that customers use this system for the purpose of supporting automation of these operations and incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This system was also not designed for incorporating into equipment for the purpose of controlling high amounts of heat (nuclear control equipment, combustion furnace control equipment, etc.).
In cases where a fault or malfunction of this product could directly threaten human life or cause an injury, its use in operations or incorporating into equipment is prohibited. If this product is used, the customer assumes all responsibility for ensuring safe operation.

Operating Procedures

This section describes the usage requirements for the TS5000 robot. Usage under different conditions could expose the user to unexpected risks. Be sure to follow the required conditions.

1. Main Specifications of Product Refer to the robot manual for your robot.
2. Usage Restrictions: Tables 1 to 6 provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application of using product (restrictions on use)	
1) Intended use	Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.). (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table1 Restricted Specification Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Press the robot by hand to move to a location where it is easy to perform work. [6] Mount the end effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power while these robot covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that a stop to wait for an operation signal during operation was an operation stop - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes
2) Electrical restrictions	- Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with Robot Manual Specifications Table Compliant with Robot Manual Specifications Table Compliant with Robot Manual Specifications Table Compliant with Robot Manual Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: The packing box shall be able to be transferred by a transportation worker. Unpacking: The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. Transfer: The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. Installation: The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching: Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if someone enters the hazard zone, they will not get entrapped by the robot or peripheral equipment. Operation: A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Repacking: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Maintenance: Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Disposal: Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.	

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period

The warranty period of the product is one of the following, whichever is shorter.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge only if this product has been operated and used in accordance with the instruction manual supplied with the product.
A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimer

The warranty does not include the following cases.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables*1).
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When a product purchased in Japan is taken abroad

1) Consumable parts refer to the maintenance replacement parts described in the target controller manual.

4. Safety Information

(1) SHIBAURA MACHINE shall provide no guarantees for the standard performance of this product when used outside the specifications.
(2) If the customer does not observe the safety information appearing for DANGER, WARNING, and CAUTION contained in this instruction manual, SHIBAURA MACHINE will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.
(3) Note that dangers, warnings, cautions, notices, and other information are those assumed by SHIBAURA MACHINE

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.
Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.
●ISO 10218, ISO 10218-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Cautions on Safety

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.

Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of Indications]

Indication	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or long-term hospital visits for treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠ DANGER

- Be sure to read the Safety Manual described in the applicable Controller Manual carefully to ensure safe transportation and disposal of the controller. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Overview

This part presents an overview of the alarms.

1. Occurrence of Alarms

An alarm occurs when an error is found in the controller or the robot operation or settings.
When an alarm occurs, a buzzer sound is emitted[1)] from the controller, and the "ALARM" LED of the controller turns on.
When this happens, if the teach pendant is connected, an alarm message appears on the screen.

1) The buzzer sound may not be emitted for certain parameter settings.

2. Confirming the Alarm Message

This describes how to confirm the alarm message.
You can confirm the TP5000 and TP1000 alarm messages.
The procedures for confirming in each teach pendant are as follows.

[1] TP5000

Tap the Alarm button in the status bar on the screen to open the alarm window.



Figure 2.1 Location of the ALARM button on the TP5000

[2] TP1000

Confirm by pressing the [ERROR] button on the TP1000.

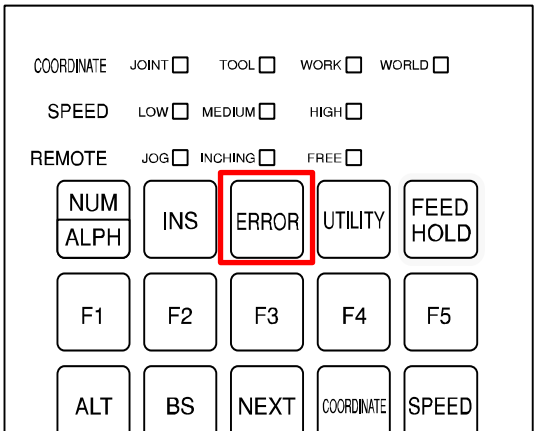


Figure 2.2 Location of the ERROR button on the TP1000

3. Reading the Alarm Message

The alarm message consists of the following parts.

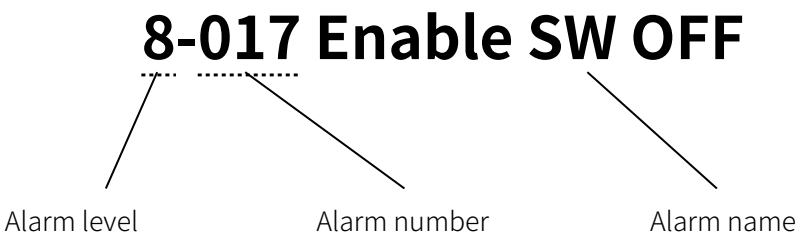


Figure 3.1 Alarm Classification

Alarm level	This shows the robot operation when the alarm occurred.
Alarm number	This is the number assigned to the specific alarm.
Alarm name	This is the name assigned to the specific alarm.

Alarms are classified into the following levels.

Level 8	The robot performs an emergency stop, and the servo is turned off.
Level 4	The robot is stopped in the current state.
Level 2	The robot is stopped just before the problem occurrence location in the SCOL2 program.
Level 1	The robot displays the alarm message only.

Details

This part describes the alarms, their causes, and corrective actions.

Possible Power-on Faults

"READY" LED does not turn on

Description	The "READY" LED in the front bottom section of the controller does not turn on even when the POWER switch[1)] is set to ON.
Cause	① The controller AC power is not being properly supplied (200 VAC). ② The "READY" LED may be faulty. ③ The internal switching power supply may be faulty.
Remedy	① Check the voltage of the controller AC power (200 VAC). ② Replace the "READY" LED. ③ Replace the switching power supply.
Remarks	Please request part replacement from the SHIBAURA MACHINE Customer Service.

1) POWER switch: Switch that turns the controller power OFF and ON

Nothing is displayed on the teach pendant.

Description	Nothing is displayed on the teach pendant.
Cause	① The teach pendant is not connected correctly. ② The teach pendant may be faulty. ③ Defective main module ④ The internal switching power supply may be faulty.
Remedy	① Check the connections for the teach pendant. ② Check for any connector or cable damage, and if there are no problems, replace the teach pendant. ③ Replace the main module. ④ Replace the internal switching power supply.
Remarks	Please request part replacement from the SHIBAURA MACHINE Customer Service. For module names, see Appendix, “Controller Module Names”.

Level 8 Alarms

This chapter describes the level 8 alarms that occur in the TS5000, their causes, and corrective actions.
When a level 8 alarm is detected, the robot performs an emergency stop, and the servo is turned off.

8-001 Undefined int

Description	Abnormal interrupt processing occurred in the main module.
Cause	① Main module failure ② Servo amplifier module failure.
Remedy	① Replace the main module. ② Replace the servo amplifier module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-004 Watchdog error

Description	A CPU periodic monitoring circuit error was detected in the main module.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-005 PLC CPU error

Description	The main module is faulty.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-007 Servo soft watchdog1 error
8-008 Servo soft watchdog2 error

Description	The main control CPU detected a servo CPU error in the mutual monitoring (software-based monitoring) of the main control and servo control.
Cause	① Main module failure ② Servo amplifier module failure.
Remedy	① Replace the main module. ② Replace the servo amplifier module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-013 AC alarm

Description	A power supply error was detected by the AC power supply monitor circuit.
Cause	The following conditions were detected in the servo power module. - The AC power supply (200 V to 250 V) was shut off for 40 msec or longer. - The AC power supply dropped to 190 V or less.
Remedy	Check the voltage and whether there was an instantaneous power outage of the controller AC power supply, and turn the power supply OFF and ON again. If no problem is found in the above, the servo amplifier module is faulty.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-014 Emergency stop SW ON

Description	The emergency stop switch was pressed.
Cause	① The emergency stop switch was pressed. ② A contactor is not making proper contact.
Remedy	① The emergency stop switch is included with the teach pendants (TP5000, TP1000). If the button has not been pressed, the switch may be faulty. ② Check the wiring. An internal connector may not be making proper contact, and so check the connections again. * If the teach pendant will not be used, connect the TP jumper connector supplied with the controller.
Remarks	Make the jumper connection if the teach pendant will not be used. For details, refer to the Interface Manual.

8-017 Enable SW OFF

Description	The enable switch is off.
Cause	① The enable switch of the teach pendant is not being gripped in MANUAL mode. ② The SAFE connector TP_ENA*A or TP_ENA*C is open.
Remedy	① Grip the enable switch of the teach pendant. ② Check the SAFE connection. (SAFE connector TP_ENA*A or TP_ENA*C)
Remarks	① The enable switch of the teach pendant is a three-level switch. For safety, it must be gripped in the middle position. ② For details, refer to the Interface Manual, "SAFE Connection".

8-018 Servo CPU error

Description	A servo CPU error was detected.
Cause	The main module is faulty.
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-019 Enable SW ON when starting

Description	When the power was turned on or when the mode was switched to MANUAL mode, the ON status of the enable switch on the teach pendant was detected.
Cause	① The enable switch on the teach pendant was already pressed. ② The enable switch of the teach pendant may have failed.
Remedy	① Set the enable switch to OFF, and then reset the alarm. ② Replace the teach pendant.
Remarks	If the power is turned on while holding down the enable switch, the controller determines that the enable switch may have failed and thus remains pressed.

8-020 Servo amplifier module model mismatch

Description	The servo amplifier module of the controller does not correspond to the set parameters.
Cause	① A wrong parameter file has been written. ② The servo amplifier module is faulty.
Remedy	① Write the parameters included in the DVD that came with the controller. Alternatively, restore the parameters to the state in which there were backed up. ② Replace the servo amplifier module.
Remarks	

8-023 PN voltage error

Description	The PN voltage detection value has become out of the specified range when starting with Axis.
Cause	The main module is faulty.
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-024 RGST error

Description	Regenerative discharge did not turn off when servo was turned on and voltage did not increase.
Cause	The servo power module is faulty.
Remedy	Replace the servo power module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-025 Servo PL WDT error

Description	A PL watch dog error was detected.
Cause	The main module is faulty.
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-027 Slow charge error

Description	Because the capacitor charge was not working normally when the servo was turned on, the servo ON operation failed. This alarm occurs when the servo supply voltage has not reached the specified voltage within a certain period of time after the servo was turned on.
Cause	① The AC power supply does not satisfy the controller input power supply specifications value (200 VAC) or is unstable. ② The voltage in the AC power supply connection connector (CN1) of the controller has dropped with respect to the equipment supply voltage. ③ The servo ON operation after servo OFF was too fast. ④ The servo power module is faulty.
Remedy	① Verify whether the controller input power supply (200 VAC) satisfies the controller input power supply specifications value and whether the voltage is stable. ② The power cable may be too thin or too long. Check the power cable diameter and cable length. (Power cable conductor thickness: 3.5 to 5.5 [mm^2]) ③ Perform the servo off/on operation more slowly. ④ Replace the servo power module or servo amplifier module.
Remarks	● The power supply voltage specifications value varies by the controller, and so refer to Controller Manual: Installation for your controller. ④ Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-030 Servo power Low volt

Description	The servo power supply (Approx. 300 VDC) inside the controller has dropped to 130 V or less.
Cause	① Drop in the controller AC power supply (200V) voltage. ② If the above is checked and no problem is found, the servo power module is faulty.
Remedy	① Verify whether the voltage of the controller input power supply is adequate and is stable. ② Replace the servo power module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-032 Servo power Over volt

Description	The servo power supply (Approx. 300 VDC) inside the controller has reached 420 V or more.
Cause	① The regenerative resistor may be damaged. ② The servo amplifier module and/or servo power module is/are faulty.
Remedy	① Replace the regenerative resistor. ② Replace the servo amplifier module and/or servo power module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-033 Distribution error

Description	A distribution processing calculation error was detected.
Cause	As a result of a movement command (such as MOVE) calculation, the robot could no longer perform movement. This alarm does not occur due to an SCOL program error. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service. At that time, please send the system software version, selected program file, and parameter file to SHIBAURA MACHINE Customer Service.
Remarks	

8-034 Unable to trans

Description	A distribution target position coordinate conversion failure error was detected.
Cause	A conflict occurred in the results of coordinate conversion in the internal calculation, and thus robot operation cannot be performed. This alarm does not occur due to an SCOL program error. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

8-035 Target position out of coordinate range

Description	Moved out of operation range during execution of CP operation.
Cause	While executing a CP operation such as MOVES or MOVEC, the target position has moved outside the robot's motion range.
Remedy	Please review the program and teaching points.
Remarks	

8-041 Axis1 exceed speed
8-042 Axis2 exceed speed
8-043 Axis3 exceed speed
8-044 Axis4 exceed speed
8-045 Axis5 exceed speed
8-046 Axis6 exceed speed
8-047 Axis7 exceed speed
8-048 Axis8 exceed speed

Description	The results of the internal calculation processing of the operation command generates a command value that exceeds the limit speed of the corresponding joint axis.
Cause	① The speed is superimposed in the same vector direction when movement is performed using a program PASS command.  Operation command in different direction　　Operation command in same or similar direction Note: A PASS command in the same rotation direction leads to superimposing of the speed commands, which results in exceeding of the speed command. ② Error in specifying TORQUE command method. ③ Or the occurrence of a servo system alarm such as 8-42X AxisX Position error.
Remedy	① Correct the program so that the PASS rate is reduced. ② Refer to the Robot Language Manual, Command Descriptions "TORQUE", and revise the program. ③ This alarm could also occur when an 8-42X AxisX Position error or other error occurs, and so check whether a servo system alarm has occurred.
Remarks	If the alarm still occurs even after taking the above corrective action, please contact SHIBAURA MACHINE Customer Service.

8-065 Main CPU peripheral failed

Description	An error was detected in the peripheral circuit control module of the main module.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-066 QSPI ROM initialization failed

Description	Initialization of the QSPI-ROM circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-067 Io-EX initialization failure

Description	Initialization of the extended IO circuitry of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-068 RTC initialization failure

Description	Initialization of the RTC circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-069 ADC initialization failure

Description	Initialization of the AD conversion circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-070 LAN initialization failure

Description	Initialization of the LAN circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-071 LAN1 PHY address illegal

Description	A PHY address error was detected for LAN1 of the main module.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-072 LAN2 PHY address illegal

Description	A PHY address error was detected for LAN 2 of the main module.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-073 LAN1 config failed

Description	Configuration of LAN1 of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-074 LAN2 config failed

Description	Configuration of LAN 2 of the main module failed.
Cause	The settings for the IP address, subnet mask, or default gateway may be incorrect.
Remedy	Check [E000], [E001], and [E002] in the ETHERNET.PAR file.
Remarks	

8-075 Communication start failed

Description	Communication start failed within the system.
Cause	There may be a fault in the system.
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-076 Io-EX0 communication failed

Description	Communication with the extended IO0 circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-077 Io-EX1 communication failed

Description	Communication with the extended IO1 circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-078 Io-EX2 communication failed

Description	Communication with the extended IO2 circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-079 RTC communication failed

Description	Communication with the RTC circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-080 ADC communication failed

Description	Communication with the AD conversion circuit of the main module failed.
Cause	Main module failure
Remedy	Replace the main module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-081 Servo task1 WDT error

Description	A watchdog error was detected in servo control task 1.
Cause	① The main module is faulty. ② There may be a fault in the system.
Remedy	Replace the main module. Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-082 Servo task2 WDT error

Description	A watchdog error was detected in servo control task 2.
Cause	① The main module is faulty. ② There may be a fault in the system.
Remedy	Replace the main module. Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-083 Servo main WDT1 error

Description	A main task watchdog error was detected in servo control task 1.
Cause	① The main module is faulty. ② There may be a fault in the system.
Remedy	Replace the main module. Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-084 Servo main WDT2 error

Description	A main task watchdog error was detected in servo control task 2.
Cause	① The main module is faulty. ② There may be a fault in the system.
Remedy	Replace the main module. Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

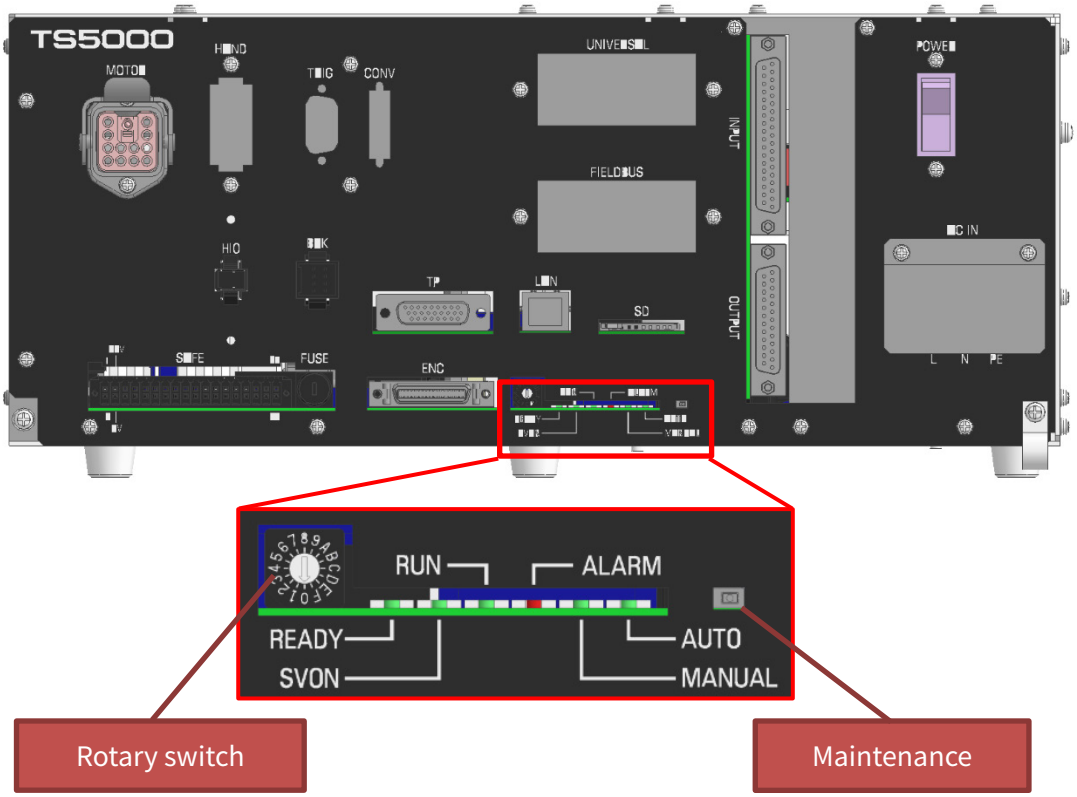
8-113 Robot parameter error

Description	A parameter error was detected in the ROBOT.PAR file.
Cause	A parameter error was detected, and so robot operation is disabled. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

8-114 Servo parameter error

Description	A parameter error was detected in the SERVO.PAR file.
Cause	A parameter error was detected, and so robot operation is disabled. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

8-115 FRAM0 space CRC error

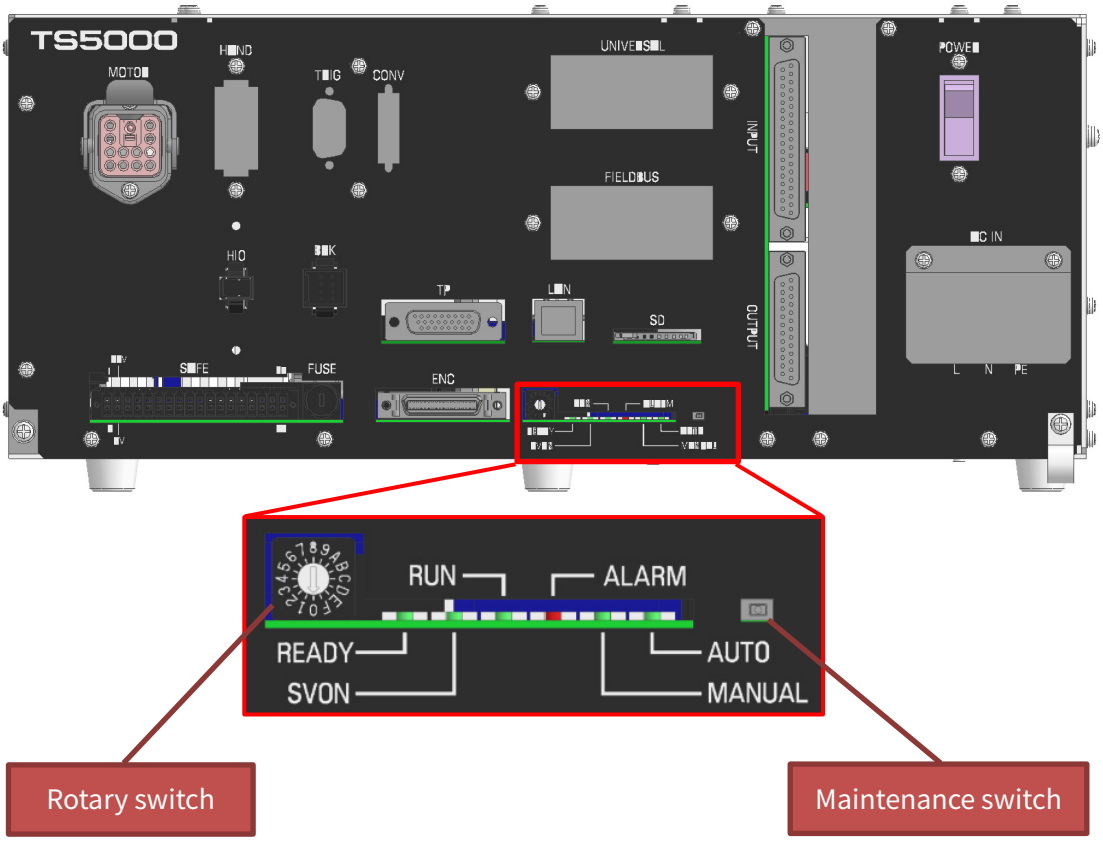
Description	A corruption of the data stored in the FRAM0 space built into the controller has been detected.
Cause	Failed to write to or read from FRAM, or base module failure.
Remedy	Replace the base module or clear FRAM contents Procedure for clearing FRAM contents 

	[Required tools] ● A stick with a fine tip such as a ballpoint pen (to press the maintenance switch) ● Flat-blade screwdriver (to turn the rotary switch) [Operating procedure] [For software version X8KCS-04A or earlier] ① Turn off the power of the controller. ② Set the rotary switch on the front panel to "B". ③ Turn on the power while pressing the maintenance switch. ④ When the READY LED on the front panel lights up, release the maintenance switch. ⑤ When FRAM is initialized correctly, "1-108 Memory erase mode" will be generated. ⑥ When the AUTO LED on the front panel lights up, the mode has switched to AUTO normally. ⑦ Set the rotary switch on the front panel to "0". [For software version X8KCS-04B or later] ① Turn off the power of the controller. ② Set the rotary switch on the front panel to "2". ③ Turn on the power while pressing the maintenance switch. ④ When the READY LED on the front panel lights up, release the maintenance switch. ⑤ When FRAM0 is initialized correctly, "1-099 FRAM0 erase mode " will be generated. ⑥ When the AUTO LED on the front panel lights up, the mode has switched to AUTO normally. ⑦ Set the rotary switch on the front panel to "0".
Remarks	[When initialized with the rotary switch set to "B"] The following items will be erased by FRAM erasing operation. ● Memory variables ➢ Data saved by the SAVEF command ➢ Data saved by the SAVEI command ➢ Data stored in MEMORYI system variables ➢ Data stored in MEMORYJ system variables ➢ Data stored in MEMORYL system variables ➢ Data stored in MEMORYP system variables ➢ Data stored in MEMORYR system variables ➢ Data stored in MEMORYT system variables ● Configuration information ➢ Selected work coordinates ➢ Selected base coordinates

	➢ Selected tool coordinates ➢ Override (when set to hold with power off in USER.PAR) ➢ Selected program file name information ➢ Breakpoint row information ● File name being edited ● Edit cursor position of the file being edited ● Error history ● Encoder pulse value for each axis ● TC mini latch register information ● Various maintenance information [When initialized with the rotary switch set to "2"] The following items will be erased by FRAM0 erasing operation. ● Configuration information ➢ Selected work coordinates ➢ Selected base coordinates ➢ Selected tool coordinates ➢ Override (when set to hold with power off in USER.PAR) ➢ Selected program file name information ➢ Breakpoint row information ● File name being edited ● Edit cursor position of the file being edited ● Error history ● Encoder pulse value for each axis ● Tcmini latch register information ● Various maintenance information

8-116 FRAM1 space CRC error

Description	A corruption of the data stored in the FRAM0 space built into the controller has been detected.
Cause	Failed to write to or read from FRAM, or base module failure.
Remedy	Replace the base module or clear FRAM contents Procedure for clearing FRAM contents



[Necessary tools]

- A stick with a fine tip such as a ballpoint pen (to press the maintenance switch)
- Flat-blade screwdriver (to turn the rotary switch)

[Operating procedure]

① Turn off the power of the controller.
② Set the rotary switch on the front panel to "B".
③ Turn on the power while pressing the maintenance switch.
④ When the READY LED on the front panel lights up, release the maintenance switch.
⑤ When FRAM is initialized correctly, "1-108 Memory erase mode" will be generated.
⑥ When the AUTO LED on the front panel lights up, the mode has switched to AUTO normally.
⑦ Set the rotary switch on the front panel to "0".

Remarks	The following items will be erased by FRAM erasing operation. ● Memory variables ➢ Data saved by the SAVEF command ➢ Data saved by the SAVEI command ➢ Data stored in MEMORYI system variables ➢ Data stored in MEMORYJ system variables ➢ Data stored in MEMORYL system variables ➢ Data stored in MEMORYP system variables ➢ Data stored in MEMORYR system variables ➢ Data stored in MEMORYT system variables ● Configuration information ➢ Selected work coordinates ➢ Selected base coordinates ➢ Selected tool coordinates ➢ Override (when set to hold with power off in USER.PAR) ➢ Selected program file name information ➢ Breakpoint row information ● File name being edited ● Edit cursor position of the file being edited ● Error history ● Encoder pulse value for each axis ● Tcmini latch register information Various maintenance information

8-117 FRAM0 space write fail

Description	Failure to write data to the FRAM0 space built into the controller has been detected.
Cause	Failed to write to or read from FRAM, or base module failure.
Remedy	Contact SHIBAURA MACHINE Customer Service, and let them know the system software version and all files in the controller, including the robot program.
Remarks	

8-129 User Alarm
8-130 User Alarm
8-131 User Alarm
8-132 User Alarm
8-133 User Alarm
8-134 User Alarm
8-135 User Alarm
8-136 User Alarm

Description	A level 8 user alarm occurred.
Cause	A level 8 user alarm occurred.
Remedy	Remove the alarm conditions that were set by the customer.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual "[U026] USER ALARM".

8-241 Controller type error

Description	A parameter error was detected in the MACHINE.PAR file.
Cause	A parameter error was detected, and so robot operation is disabled. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

8-242 CUNET failure

Description	Extended I/O module failure was detected.
Cause	① Main module failure ② Base module failure
Remedy	① Replace the main module. ② Replace the base module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-243 System I/O module failure

Description	System I/O module failure was detected.
Cause	The system I/O module failure
Remedy	Replace the system I/O module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-244 Hand 1 module failure

Description	Hand 1 module failure was detected.
Cause	Hand 1 module failure
Remedy	Replace the hand 1 module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-245 Hand 2 module failure

Description	Hand 2 module failure was detected.
Cause	Hand 2 module failure
Remedy	Replace the hand 2 module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-246 Option1 I/O module failure

Description	Option1 I/O module failure was detected.
Cause	Option1 I/O module failure
Remedy	Replace the option1 I/O module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-249 Failure of cable for option3 conveyor synchronization

Description	Option3 conveyor option could not be detected.
Cause	The cable for option3 conveyor synchronization is faulty.
Remedy	Replace the cable for option3 conveyor synchronization.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-250 Option4 type error

Description	A parameter error was detected in the MACHINE.PAR file.
Cause	A parameter error was detected, and so robot operation is disabled. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

8-251 Option4 AD module failure

Description	Option4 AD module failure was detected.
Cause	Option4 AD module failure
Remedy	Replace the option4 AD module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-252 Option4 I/O module failure

Description	Option4 I/O module failure was detected.
Cause	Option4 I/O module failure
Remedy	Replace the option4 I/O module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-258 Backup memory error

Description	The backup memory failed.
Cause	Base module failure
Remedy	Replace the base module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module. For module names, see Appendix, “Controller Module Names”.

8-269 PLC user alarm
8-270 PLC user alarm
8-271 PLC user alarm
8-272 PLC user alarm

Description	A level 8 user alarm has occurred in the internal PLC.
Cause	A level 8 user alarm has occurred in the internal PLC.
Remedy	The user must remove the alarm conditions that were set in the internal PLC.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual "[P004] PLC ALARM".

8-273 I/O fuse broken

Description	A fuse error was detected.
Cause	A fuse is blown.
Remedy	Replace the fuse. Check that the load does not exceed the specifications value of 3 (A). Also, check that the external I/O wiring is not connected in a short-circuit.
Remarks	For details on fuse replacement, refer to the Controller Manual, "Fuse Replacement".

8-274 Enable SW error

Description	An error occurred due to the enable switch.
Cause	① The two contact states of the teach pendant enable switch do not match. ② The power was turned on while the enable switch was being grasped. ③ Teach pendant enable switch state mismatch has occurred.
Remedy	① Turn the power on again. If the alarm still appears, the teach pendant switch or wiring may be faulty. ② Release the enable switch, and turn the power on again. ③ Check the SAFE connection circuit. (SAFE connector TP_ENA*A or TP_ENA*C)
Remarks	Please contact SHIBAURA MACHINE Customer Service.

8-275 EMG SW error

Description	An error was detected in the emergency stop switch.
Cause	The two (2) circuits for the external emergency stop contact do not match.
Remedy	① There may be a fault in the emergency stop wiring or emergency stop switches, and so refer to 8-014 Emergency stop SW ON, and check that there are no problems with the emergency stop switch circuit. ② When a switch state mismatch is detected, the alarm is retained, and so turn the power off and then on again to reset the alarm.
Remarks	

8-277 PLC stop

Description	The internal PLC ladder program is stopped.
Cause	① The ladder program was stopped by TC-WORX[1]. ② An alarm occurred in the PLC, and the ladder program was stopped.
Remedy	① Run the ladder program using TC-WORX. ② Turn the power on again. Alarms that occur when the ladder program was intentionally stopped are normal. Either turn the power off and then on again, or send the start command from the PLC programmer to rerun the ladder program.
Remarks	TC-WORX is a Tcmini ladder program development tool that runs on a PC. For details on installation, refer to the STA1147_TC-WORX Setup Manual. For details on the operating procedures, refer to STA1148_TC-WORX Introductory Manual.

8-278 Ext DC24V regulator error

Description	An error occurred in the external 24 (V) power supply.
Cause	The output of the I/O external power supply provided by the user has dropped.
Remedy	Check the I/O power supply output provided by the customer.
Remarks	

8-279 Ext resistor overheat

Description	The temperature of the regenerative resistor exceeds 130 (C).
Cause	Turn off the power, and let the regenerative resistor cool down. The following are possible causes of overheating of the resistor. ① The robot is overloaded. ② Appropriate acceleration/deceleration is not set for the load. ③ The robot has collided with an object. ④ There is a problem in the robot drive system (motor, reduction gears, or other parts). ⑤ The servo power module failure.
Remedy	① Use the PAYLOAD command to adjust the acceleration/deceleration and servo gain. ② Use the PAYLOAD, ACCEL, and DECEL commands to adjust to a suitable acceleration/deceleration. PAYLOAD = {<Weight>, <Offset of center of gravity>} The value for <Weight> is the weight of the load applied to the hand tip of the robot, and it is specified in kg units. The value for <Center of gravity offset> is how far the weight of the load applied to the robot hand tip is from the tool center of the hand tip, and it is specified in mm units. ACCEL = <expression>, DECEL = <expression> These are specified as percentages of the standard acceleration. ACCEL is used for acceleration, and DECEL is used for deceleration. For details, refer to the Robot Language Manual. ③ Remove the cause of the interference. ④ Move the robot by hand, and check that there is no catching, rattling, or other problems. ⑤ Replace the servo amplifier module and/or servo power module.
Remarks	If module replacement is required, please contact SHIBAURA MACHINE Customer Service. For module names, see Appendix, “Controller Module Names”.

8-283 Illegal master mode key

Description	When the key switch is used to change the master mode, master mode mismatch occurred.
Cause	When the teach pendant displayed other than the top screen, the key switch was used to change the master mode.
Remedy	① Return the teach pendant to the top screen. ② Return the key switch to the original master mode prior to change.
Remarks	

8-284 Controller mode SW error

Description	A mismatch in the mode switch input circuit of the teach pendant was detected.
Cause	The states of the two contacts in the teach pendant mode switch do not match.
Remedy	Turn the power on again. If the alarm still appears, the teach pendant mode switch or wiring may be faulty.
Remarks	

8-285 Master mode mismatch

Description	A mismatch between the master mode of the teach pendant TP5000 and the master mode inside the controller was detected.
Cause	The master mode of the teach pendant TP5000 and the master mode inside the controller do not match
Remedy	Change the master mode of the teach pendant TP5000.
Remarks	

8-354 Fieldbus Offline

Description	An error occurred in fieldbus communication, causing the device to go offline.
Cause	A fieldbus communication error occurred because the settings of the fieldbus parameters (communication speed, node number) do not match with those on the master side, resulting in the device going offline.
Remedy	Investigate the cause with the fieldbus master device.
Remarks	

8-355 Fieldbus module error

Description	The type of the fieldbus module is incorrect.
Cause	① No fieldbus module is installed. ② The type of the fieldbus module of the controller model and the type of the fieldbus module installed are different.
Remedy	① Install fieldbus module that matches with the controller model. ② Check the controller model and fieldbus module type, and change the fieldbus module if they do not match.
Remarks	If module replacement is required, please contact SHIBAURA MACHINE Customer Service. For module names, see Appendix, “Controller Module Names”.

8-361 IP1 status alarm
8-362 IP2 status alarm
8-363 IP3 status alarm
8-364 IP4 status alarm
8-365 IP5 status alarm
8-366 IP6 status alarm
8-367 IP7 status alarm
8-368 IP8 status alarm

Description	The Ethernet connection was disconnected during running of the program.
Cause	The Ethernet connection was disconnected.
Remedy	Check the connection between the robot controller and the device connected via Ethernet. Check that the Ethernet cable is not damaged.
Remarks	The occurrence and level for this alarm can be set in the ETHERNET.PAR file. For details, refer to the Parameter Manual.

8-411 Axis1 failed to read origin information
8-412 Axis2 failed to read origin information
8-413 Axis3 failed to read origin information
8-414 Axis4 failed to read origin information
8-415 Axis5 failed to read origin information
8-416 Axis6 failed to read origin information
8-417 Axis7 failed to read origin information
8-418 Axis8 failed to read origin information

Description	Failed to load the origin information of the motor of each joint axis.
Cause	Information cannot be read from the encoder.
Remedy	If this happens, restart the controller. If this error occurs frequently, replace the motor of the error causing joint axis.
Remarks	

8-421 Axis1 position error
8-422 Axis2 position error
8-423 Axis3 position error
8-424 Axis4 position error
8-425 Axis5 position error
8-426 Axis6 position error
8-427 Axis7 position error
8-428 Axis8 position error

Description	The difference between the command position and actual position for the motor of the target joint axis exceeded the specified value.
Cause	① The robot cannot move due to an external cause such as interference (collision). ② The reduction gears of the target joint axis or the ball screw spline may have failed. (For the 1st, 2nd, and 4th axes, the reduction gear may be faulty. For the 3rd axis, the ball screw spline may be bent.) ③ The torque value that was set by the TORQUE command is too small. ④ The motor or servo amplifier module of the relevant joint axis is faulty.
Remedy	① Remove any causes of interference or other external factors. ② Set to servo-free mode, and move the target joint axis by hand, and if there is any catching or the movement is not smooth, the reduction gear may be damaged, or the ball screw spline may be bent. Replace the part in the applicable location. ③ Set the TORQUE value again to a value that enables full movement by the robot. ④ Replace the motor and servo amplifier module of the relevant joint axis. * To determine whether the motor is faulty or normal, measure the resistances between the phases of the motor wires. Normally, the resistances between R-S, S-T, and T-R are all the same. If any of these is different, the motor has failed.
Remarks	If module replacement is required, please contact SHIBAURA MACHINE Customer Service.

8-451 Axis1 motor locked
8-452 Axis2 motor locked
8-453 Axis3 motor locked
8-454 Axis4 motor locked
8-455 Axis5 motor locked
8-456 Axis6 motor locked
8-457 Axis7 motor locked
8-458 Axis8 motor locked

Description	The motor does not rotate even when the maximum torque is output. This alarm does not occur if the torque is limited by the TORQUE command.
Cause	① The robot cannot move due to an external cause such as interference (collision). ② If "8-48X AxisX IPM error" also occurs, the motor line may be short-circuited or disconnected, or the servo amplifier module or motor of the relevant joint axis may be faulty. ③ The reduction gears of the target joint axis or the ball screw spline may have failed.
Remedy	① Review the program, check if there is possibility of interference, and remove any external cause of interference. ② Check if there is any looseness in the motor line connector, or if the cable is damaged. If there is no abnormality in the motor line, replace the motor or servo amplifier module. * To determine if the motor is faulty, measure the resistances between phases of the motor line. Normally, the resistances between R-S, S-T, and T-R are all the same. If any of these is different, the motor has failed. ③ Set to servo-free mode, and move the target joint axis by hand, and if there is any catching or the movement is not smooth, the reduction gear may be damaged, or the ball screw spline may be bent. Replace the faulty parts.
Remarks	If the alarm does not clear after taking the above measures, please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module.

8-461 Axis1 motor speed over
8-462 Axis2 motor speed over
8-463 Axis3 motor speed over
8-464 Axis4 motor speed over
8-465 Axis5 motor speed over
8-466 Axis6 motor speed over
8-467 Axis7 motor speed over
8-468 Axis8 motor speed over

Description	Motor rotation has exceeded the specified value (maximum rotation speed for the respective joint axis).
Cause	① When the program operation command makes a PASS connection, the speed is superimposed in the same vector direction.  Operation command in different direction　　Operation command in same or similar direction ② There is an error in the TORQUE command specifying method. ③ When the CN2 connector (motor cable connector) was removed while the power was off, if strong vibrations were applied to the robot or it was moved quickly (approx. 6000 rpm or more) by hand, this alarm may occur when the power is turned on. ④ This can occur due to hand interference or mechanical causes such as drive system problems.
Remedy	① Correct the program so that the PASS rate is reduced. ② Correct the program by referring to "Command Descriptions 'TORQUE'" in the Robot Language Manual. ③ Perform the recovery work by following the recovery procedure in the Appendix "Recovery Procedure for Origin Position Data" in the Maintenance Manual. ④ Remove any mechanical causes of the error.
Remarks	

8-481 Axis1 IPM error
8-482 Axis2 IPM error
8-483 Axis3 IPM error
8-484 Axis4 IPM error
8-485 Axis5 IPM error
8-486 Axis6 IPM error
8-487 Axis7 IPM error
8-488 Axis8 IPM error

Description	An error was detected in the motor drive transistor on AxisX servo amplifier module. Overcurrent: The power cable of the target joint axis motor may have short-circuited. Overheat: The temperature of the transistor on the servo amplifier module is above 110°C. Supply voltage drop: The supply voltage to the transistor of the servo amplifier module has dropped below 12V. (normal voltage:15 [V])
Cause	① Appropriate acceleration/deceleration is not set for the load. (The load on the robot is too high.) <Overheat> ② The target joint axis motor cable may have short-circuited. <Overcurrent> ③ The motor of the target joint axis may have failed. ④ The servo amplifier module of the relevant joint axis is faulty.
Remedy	① Turn off the power, and let the motor drive transistor cool down. Then, review the program, and adjust to an appropriate acceleration/deceleration either by using the ACCEL, DECEL, and other acceleration/deceleration setting commands or by using the PAYLOAD command. ② Identify the location with the poor contact or short-circuit by referring to the Controller Manual: Installation. If there is a broken wire, replace the cable for the relevant location. ③ To determine if the motor is faulty, measure the resistances between phases of the motor line. Normally, the resistances between R-S, S-T, and T-R are all the same. If any of these is different, the motor has failed. Replace the motor. ④ Replace the servo module of the target joint axis.
Remarks	* If this problem occurs frequently and the situation does not improve, the servo amplifier module or the corresponding joint axis motor may be faulty, or the motor power line in the vicinity may be disconnected. Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module.

8-491 Axis1 electric thermal
8-492 Axis2 electric thermal
8-493 Axis3 electric thermal
8-494 Axis4 electric thermal
8-495 Axis5 electric thermal
8-496 Axis6 electric thermal
8-497 Axis7 electric thermal
8-498 Axis8 electric thermal

Description	If the load for the target joint axis motor was large, and a current at the motor rated value or higher continued to flow, this alarm occurs to protect the motor. (This alarm is monitored by software.)
Cause	① The robot cannot move due to an external cause such as interference (collision). ② The reduction gears of the target joint axis or the ball screw spline may have failed. (For the 1st, 2nd, and 4th axes, the reduction gear may be faulty. For the 3rd axis, the ball screw spline may be bent.) ③ The motor of the target joint axis may have failed. ④ The servo amplifier module is faulty. ⑤ The servo module may have failed.
Remedy	① Remove the cause of the interference or other external cause. ② Set to servo-free mode, and move the target joint axis by hand, and if there is any catching or the movement is not smooth, the reduction gear may be damaged, or the ball screw spline may be bent. Replace the part in the applicable location. ③ Replace the motor and servo amplifier module of the relevant joint axis. * To determine if the motor is faulty, measure the resistances between phases of the motor line. Normally, the resistances between R-S, S-T, and T-R are all the same. If any of these is different, the motor has failed.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module.

8-511 Axis1 current feedback
8-512 Axis2 current feedback
8-513 Axis3 current feedback
8-514 Axis4 current feedback
8-515 Axis5 current feedback
8-516 Axis6 current feedback
8-517 Axis7 current feedback
8-518 Axis8 current feedback

Description	The current feedback value just before the servo was turned on is abnormal.
Cause	The servo amplifier module may be faulty.
Remedy	Replace the servo amplifier module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module.

8-521 Axis1 servo parameter
8-522 Axis2 servo parameter
8-523 Axis3 servo parameter
8-524 Axis4 servo parameter
8-525 Axis5 servo parameter
8-526 Axis6 servo parameter
8-527 Axis7 servo parameter
8-528 Axis8 servo parameter

Description	An execution parameter calculation error was detected in AxisX
Cause	① The servo execution parameters could not be calculated because the encoder could not be recognized. ② A value outside the setting range was set for the servo parameter.
Remedy	① If this error occurs together with 8-57X AxisX encoder type error, refer to 8-57*, and clear the encoder-related alarm. ② Contact SHIBAURA MACHINE Customer Service.
Remarks	If 8-57X AxisX encoder type error has occurred, the execution parameter cannot be calculated based on the encoder type, and so this alarm also occurs at the same time.

8-531 Axis1 over current
8-532 Axis2 over current
8-533 Axis3 over current
8-534 Axis4 over current
8-535 Axis5 over current
8-536 Axis6 over current
8-537 Axis7 over current
8-538 Axis8 over current

Description	Overcurrent was detected in AxisX servo amplifier. (It was detected by the software that current greater than the detection value set in the parameter flowed.)
Cause	① Appropriate acceleration/deceleration is not set for the load. (The load on the robot is too high.) ② The target joint axis motor cable may have short-circuited. <Overcurrent> ③ The motor of the target joint axis may have failed. ④ The servo amplifier module of the relevant joint axis is faulty.
Remedy	① Review the program, and adjust to appropriate acceleration/deceleration either by using the ACCEL, DECEL, and other acceleration/deceleration setting commands or by using the PAYLOAD command. ② Identify the location with poor contact or short-circuit by referring to the Controller Manual: Installation. Replace the cable at the problem location if it is damaged. ③ To determine whether the motor is faulty or normal, measure the resistances between the phases of the motor wires. Normally, the resistances between R-S, S-T, and T-R are all the same. If any of these is different, the motor has failed. Replace the motor. ④ Replace the servo amplifier of the target joint axis.
Remarks	* If this problem occurs frequently and the situation does not improve, the servo amplifier module or the corresponding joint axis motor may be faulty, or the motor power line in the vicinity may be short-circuited. Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module.

8-541 Axis1 amp rotary switch error
8-542 Axis2 amp rotary switch error
8-543 Axis3 amp rotary switch error
8-544 Axis4 amp rotary switch error
8-545 Axis5 amp rotary switch error
8-546 Axis6 amp rotary switch error
8-547 Axis7 amp rotary switch error
8-548 Axis8 amp rotary switch error

Description	The amplifier module switch setting for the target joint axis is incorrect.
Cause	① No amplifier module was set to the target joint axis. ② Multiple amplifier modules were set to the target joint axis.
Remedy	① Check whether the amplifier that was set to the target joint axis is inside the controller. If there is an amplifier module for the target joint axis, the amplifier may have failed. Replace it. ② Check the rotary switch of the amplifier again, and correct the setting of the rotary switch.
Remarks	

8-561 Axis1 communication error
8-562 Axis2 communication error
8-563 Axis3 communication error
8-564 Axis4 communication error
8-565 Axis5 communication error
8-566 Axis6 communication error
8-567 Axis7 communication error
8-568 Axis8 communication error

Description	An encoder broken wire error was detected in AxisX.
Cause	① The encoder line has a broken wire. ② The encoder may have failed. ③ The servo amplifier module may be faulty.
Remedy	① Replace the encoder line. ② Replace the encoder (motor). ③ Replace the servo amplifier module.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module or encoder.

8-571 Axis1 encoder type error
8-572 Axis2 encoder type error
8-573 Axis3 encoder type error
8-574 Axis4 encoder type error
8-575 Axis5 encoder type error
8-576 Axis6 encoder type error
8-577 Axis7 encoder type error
8-578 Axis8 encoder type error

Description	Encoder communication was cut off at AxisX, and the format cannot be obtained.
Cause	① The encoder line has a broken wire. ② The encoder may have failed. ③ The servo amplifier module may be faulty.
Remedy	① Replace the encoder line. ② Replace the servo amplifier module. ③ Replace the encoder.
Remarks	Please contact SHIBAURA MACHINE Customer Service as it may be necessary to replace the module or encoder. If this alarm occurs, 8-52X AxisX servo parameter also occur.

8-581 Axis1 Enc over speed
8-582 Axis2 Enc over speed
8-583 Axis3 Enc over speed
8-584 Axis4 Enc over speed
8-585 Axis5 Enc over speed
8-586 Axis6 Enc over speed
8-587 Axis7 Enc over speed
8-588 Axis8 Enc over speed

Description	The motor rotations exceeded the specified value (encoder detection limit).
Cause	① The motor rotations exceeded the specified value (encoder detection limit). ② When the encoder is driven by the battery, the tracking speed of the internal circuit was exceeded, and the encoder was rotated (the arm was moved significantly by hand while the power was shut off). ③ If this alarm occurs together with 8-61X AxisX Enc battery empty, it is highly likely that a counter skip has occurred due to low encoder battery voltage. ④ The encoder has failed.
Remedy	① If this alarm occurs together with 8-46X AxisX motor speed over, it is highly likely that the motor rotation has actually exceeded the specified value due to interference with a workpiece or other cause. After turning the power off and on, check that the origin position is correct. ② The encoder detection limit is low when the power is turned off. Be careful that the arm is not moved a significant amount by hand when the power is OFF. After turning the power off and on, check that the origin position is correct. ③ Replace the encoder battery. After turning the power off and on, check that the origin position is correct. ④ Replace the motor.
Remarks	The encoder reset operation must be performed to clear the alarm. Because the origin position may have shifted, please contact SHIBAURA MACHINE Customer Service.

8-591 Axis1 Enc single count err
8-592 Axis2 Enc single count err
8-593 Axis3 Enc single count err
8-594 Axis4 Enc single count err
8-595 Axis5 Enc single count err
8-596 Axis6 Enc single count err
8-597 Axis7 Enc single count err
8-598 Axis8 Enc single count err

Description	A single turn count error occurred in AxisX.
Cause	The encoder has failed.
Remedy	Replace the motor.
Remarks	The motor must be replaced. Contact SHIBAURA MACHINE Customer Service. The single counter is a counter that shows the encoder angle.

8-601 Axis1 Enc multi count err
8-602 Axis2 Enc multi count err
8-603 Axis3 Enc multi count err
8-604 Axis4 Enc multi count err
8-605 Axis5 Enc multi count err
8-606 Axis6 Enc multi count err
8-607 Axis7 Enc multi count err
8-608 Axis8 Enc multi count err

Description	An encoder multi-turn counter error occurred in AxisX.
Cause	① If this alarm occurs together with 8-61X AxisX Enc battery empty, it is highly likely that a counter skip has occurred due to low encoder battery voltage. ② The encoder rotated at 6000 rpm or more because external force was applied to the arm when the power was off. ③ The encoder rotated at acceleration of 4000 rad/s^2 or more because external force was applied to the arm when the power was off. ④ The encoder may have failed.
Remedy	① Replace the encoder battery. After turning the power off and on, check that the origin position is correct. ②③ Check whether or not external force is applied to the arm. ④ Replace the motor.
Remarks	The encoder reset operation must be performed to clear the alarm. The origin may be misaligned, and so contact SHIBAURA MACHINE Customer Service.

8-611 Axis1 Enc battery empty
8-612 Axis2 Enc battery empty
8-613 Axis3 Enc battery empty
8-614 Axis4 Enc battery empty
8-615 Axis5 Enc battery empty
8-616 Axis6 Enc battery empty
8-617 Axis7 Enc battery empty
8-618 Axis8 Enc battery empty

Description	The battery voltage in AxisX is less than the specified value.
Cause	The encoder battery level is low.
Remedy	Replace the encoder battery. After turning the power off and on, check that the origin position is correct.
Remarks	The encoder reset operation must be performed to clear the alarm. The origin may be misaligned, and so contact SHIBAURA MACHINE Customer Service.

8-721 Axis1 abnormal U-phase current offset value
8-722 Axis2 abnormal U-phase current offset value
8-723 Axis3 abnormal U-phase current offset value
8-724 Axis4 abnormal U-phase current offset value
8-725 Axis5 abnormal U-phase current offset value
8-726 Axis6 abnormal U-phase current offset value
8-727 Axis7 abnormal U-phase current offset value
8-728 Axis8 abnormal U-phase current offset value

Description	The U-phase current feedback value has become out of the specified range when starting with AxisX.
Cause	① The electric charge remained when the power was turned off last time. ② The servo amplifier module is faulty.
Remedy	① Reset the alarm. ② Replace the servo amplifier module.
Remarks	

8-731 Axis1 abnormal V-phase current offset value
8-732 Axis2 abnormal V-phase current offset value
8-733 Axis3 abnormal V-phase current offset value
8-734 Axis4 abnormal V-phase current offset value
8-735 Axis5 abnormal V-phase current offset value
8-736 Axis6 abnormal V-phase current offset value
8-737 Axis7 abnormal V-phase current offset value
8-738 Axis8 abnormal V-phase current offset value

Description	The V-phase current feedback value has become out of the specified range when starting with AxisX.
Cause	① The electric charge remained when the power was turned off last time. ② The servo amplifier module is faulty.
Remedy	① Reset the alarm. ② Replace the servo amplifier module.
Remarks	

8-881	Axis1 + soft limit
8-882	Axis2 + soft limit
8-883	Axis3 + soft limit
8-884	Axis4 + soft limit
8-885	Axis5 + soft limit
8-886	Axis6 + soft limit
8-887	Axis7 + soft limit
8-888	Axis8 + soft limit

Description	The robot has moved out of the software limit when the torque was limited.
Cause	The robot fell because it could not support its own weight.
Remedy	Increase the torque value.
Remarks	

8-891	Axis1 - soft limit
8-892	Axis2 - soft limit
8-893	Axis3 - soft limit
8-894	Axis4 - soft limit
8-895	Axis5 - soft limit
8-896	Axis6 - soft limit
8-897	Axis7 - soft limit
8-898	Axis8 - soft limit

Description	The robot has moved out of the software limit when the torque was limited.
Cause	The robot fell because it could not support its own weight.
Remedy	Increase the torque value.
Remarks	

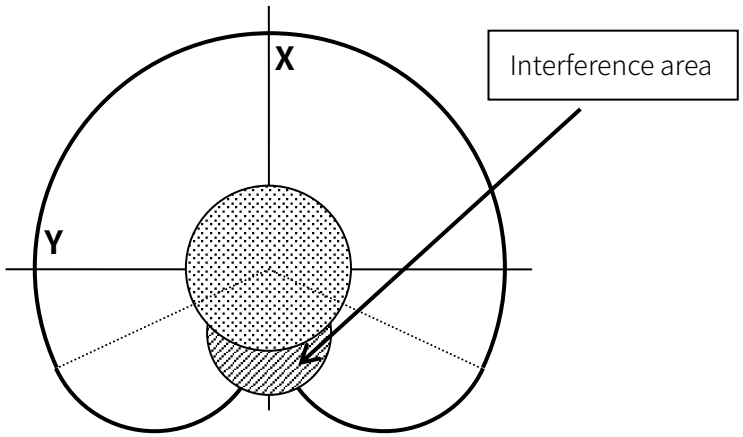
8-901	Axis1 position backup error
8-902	Axis2 position backup error
8-903	Axis3 position backup error
8-904	Axis4 position backup error
8-905	Axis5 position backup error
8-906	Axis6 position backup error
8-907	Axis7 position backup error
8-908	Axis8 position backup error

Description	The encoder value has changed by one motor rotation or more compared to the state when the power was previously turned off.
Cause	① The joint axis angle was changed while the controller power was shut off. ② The combination of the robot and the controller may be wrong. ③ The robot origin position information may be lost.
Remedy	① Check the robot origin position, and if there is no problem, perform normal operation. ② Check the combination with the controller. If it is wrong, change to the correct combination. ③ If 8-611 Axis1 Enc battery empty also occurs at the same time, the origin position information may have been lost. After replacing the battery, set the origin position again.
Remarks	This alarm no longer appears when the controller power is turned off and then on again.

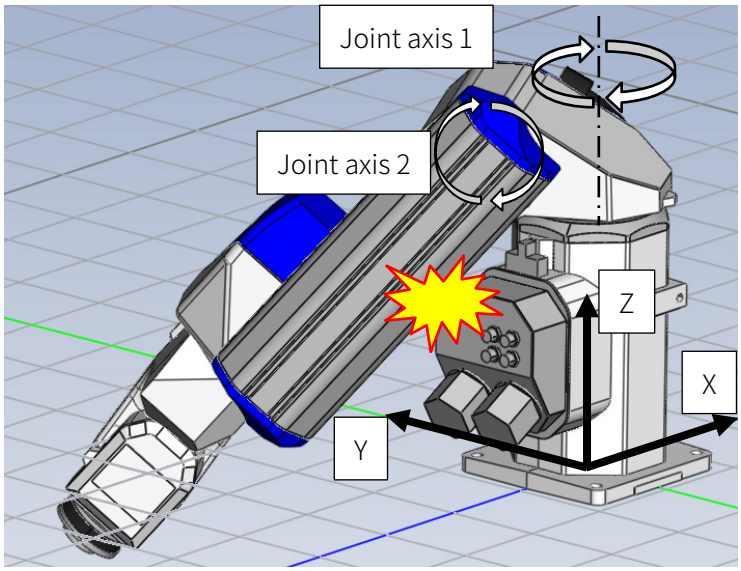
Level 4 Alarms

This chapter describes the level 4 alarms that occur in the TS5000, their causes, and corrective actions.
When a level 4 alarm is detected, the robot is stopped in the current state.

4-033 Interfere error

Description	The robot position entered an interference area.
Cause	① The robot was guided by hand into the interference area. ② In automatic operation, the linear interpolation or circular interpolation paths pass through the interference area.
Remedy	① Select a jog operation in manual guiding, and guide the target joint axis in the minus (-) direction if it is at the plus (+) operation limit, and guide the target joint axis in the plus (+) direction if it is at the minus (-) operation limit. ② Revise the program (teaching points).
Remarks	The alarm is automatically cleared when the robot moves outside the interference area. An example of the interference area of a SCARA robot is shown in the figure below. An interference area is set to prevent interference with the robot body. 

4-034 Movement limitation area

Description	The robot entered the area where joint movement is limited.
Cause	The "movement limitation area monitoring function" is provided to prevent interference with the robot (the base, arm, cable, etc.) which may occur, depending on the movement of different joints in combination. More than one movement limitation area is preset for each robot, with respect to the locations where interference may occur depending on the combined movement.  Example of interference between the arm and the robot body due to combined movement of axes No. 1 and No. 2 ***MEMO*** *When the positions of both joint axes have entered the movement limitation area, this alarm occurs, and the robot decelerates to a stop.*
Remedy	The axes can be moved away from the joint movement limitation area using the manual guide button. Clear the alarm by either moving in the opposite direction from the movement direction just before the alarm occurred, or manually guiding in the direction away from the joint movement limitation area. The alarm is automatically cleared once the axes are away from the area. (Simply resetting the alarm will not move the axes away from the area.)
Remarks	This alarm occurs in vertically articulated robots only.

4-035 Outside limitation area

Description	The robot has moved outside the limitation area.
Cause	The robot has moved outside the limitation area set in [A001] of AREA.PAR.
Remedy	The joint axes can be returned from outside the limitation area using the manual guide button. Move the joint axes in the opposite direction from the movement direction just before alarm occurrence. The alarm is automatically cleared once the axes are away from the area. (Simply resetting the alarm will not move the axes away from the area.)
Remarks	

4-036 No entry area

Description	The robot has entered the prohibited area.
Cause	The robot has moved into the prohibited area set in [A002] of AREA.PAR.
Remedy	The joint axes can be moved away from the prohibited area using the manual guide button. Move the joint axes in the opposite direction from the movement direction just before alarm occurrence. The alarm is automatically cleared once the axes are away from the area. (Simply resetting the alarm will not move the axes away from the area.)
Remarks	

4-053 VSCALBIN parameter error

Description	An argument setting in the VSCALBIN command (vision sensor coordinate system → world coordinate system conversion command) is incorrect.
Cause	① The set value is outside the reference value range (1-20) for the vision calibration table. ② The setting of the mode variable is wrong.
Remedy	① The vision calibration table numbers are 1 to 20. Set the vision calibration table number (1-20) you want to select in VISION.PAR [V00] Coordinate transformation parameter for 2D vision calibration. ② Set the correct posture variable in the world coordinate system POINT type data that is output to VSCALBOUT. [With SCARA robot] Set any of 0 to 2. 0：FREE 1：LEFTY 2：RIGHTY
Remarks	

4-065 IP1 status alarm
4-066 IP2 status alarm
4-067 IP3 status alarm
4-068 IP4 status alarm
4-069 IP5 status alarm
4-070 IP6 status alarm
4-071 IP7 status alarm
4-072 IP8 status alarm

Description	The Ethernet connection was disconnected during running of the program.
Cause	The Ethernet connection was disconnected.
Remedy	Check the connection between the robot controller and the device connected via Ethernet. Check that the Ethernet cable is not damaged.
Remarks	The occurrence and level for this alarm can be set using the parameter settings. For details, refer to the Parameter Manual “[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8”.

4-077 PLC user alarm
4-078 PLC user alarm
4-079 PLC user alarm
4-080 PLC user alarm

Description	A level 4 user alarm has occurred in the internal PLC.
Cause	A level 4 user alarm has occurred in the internal PLC. The message can be defined by the user parameters.
Remedy	The user must remove the alarm conditions that were set in the internal PLC.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual “[P004] PLC ALARM”.

4-129 User alarm
4-130 User alarm
4-131 User alarm
4-132 User alarm
4-133 User alarm
4-134 User alarm
4-135 User alarm
4-136 User alarm

Description	A level 4 user alarm occurred.
Cause	A level 4 user alarm occurred.
Remedy	Remove the alarm conditions that were set by the customer.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual "[U026] USER ALARM".

4-421	Axis1 + soft limit
4-422	Axis2 + soft limit
4-423	Axis3 + soft limit
4-424	Axis4 + soft limit
4-425	Axis5 + soft limit
4-426	Axis6 + soft limit
4-427	Axis7 + soft limit
4-428	Axis8 + soft limit
4-431	Axis1 - soft limit
4-432	Axis2 - soft limit
4-433	Axis3 - soft limit
4-434	Axis4 - soft limit
4-435	Axis5 - soft limit
4-436	Axis6 - soft limit
4-437	Axis7 - soft limit
4-438	Axis8 - soft limit

Description	The soft stroke limit was exceeded.
Cause	The stroke limit of the joint axis that was set in the software limit was exceeded.
Remedy	Select a jog operation in manual guiding, and guide the target joint axis in the minus (-) direction if it is at the plus (+) operation limit, and guide the target joint axis in the plus (+) direction if it is at the minus (-) operation limit. The alarm is automatically cleared when in the robot movement range. If this error occurs during automatic operation, the robot movement path is passing outside the soft limit. Correct the program.
Remarks	For details on the joint limit settings (J-LIMIT), see the Operator's Manual.

Level 2 Alarms

This chapter describes the level 2 alarms that occur in the TS5000, their causes, and corrective actions. When a level 2 alarm is detected, the robot is stopped just before the problem occurrence location in the SCOL2 program.

2-017 Interpreter error

Description	A program execution error was detected.
Cause	This alarm does not occur due to an SCOL program error. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

2-018 SVM error

Description	Invalid processing occurred in the internal program during program execution.
Cause	Invalid processing occurred in the internal calculation during execution of the SCOL program (ON-DO command).
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

2-019 Direct program error

Description	The direct program (program execution by DO****, manual guidance operation) cannot be executed.
Cause	Invalid processing occurred in the internal calculation during direct execution or manual guidance execution. Check that there are no errors in the operating procedure. (Refer to "Direct Execution" in the Operator's Manual.) If there is nothing wrong with the operating procedure, there may be fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and conditions under which the error occurred.
Remarks	

2-020 Unable to start

Description	The selected program cannot be executed.
Cause	If there is no error in the selected program, there may be a system fault.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and program.
Remarks	

2-021 Unable to continue

Description	The selected program was unable to be stopped or restarted.
Cause	Invalid processing occurred in the internal calculation during the stop or restart operations. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version, conditions when the error occurred including the line number, program where the error occurred, and all files in the controller.
Remarks	

2-022 Program counter error

Description	The line number of the executed program has an invalid value, and so program execution cannot be continued.
Cause	There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version, conditions when the error occurred including the line number, program where the error occurred, and all files in the controller.
Remarks	

2-023 Operation code error

Description	An ON command was issued in the subtask.
Cause	An ON condition command cannot be executed in the subtask.
Remedy	Remove the ON condition command from the subtask program of the SCOL program.
Remarks	

2-024 Operand error

Description	Invalid processing occurred in the internal calculation of the program for system variable operation.
Cause	There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version, conditions when the error occurred including the line number, program where the error occurred, and all files in the controller.
Remarks	

2-025 Invalid command

Description	A currently-invalid command was executed.
Cause	① A conveyor command was executed in a system without a conveyor synchronization option. ② An operation command was executed in a subtask. ③ A task command was executed in the ON-DO statement. ④ If auto variable is used in the ON to DO statement command, this alarm could occur depending on the timing when the ON conditions are met. Example PROGRAM MAIN ~ ON DIN(1) DO LOCAL＝1 SUB ~ END PROGRAM SUB ~ When the ON condition is met during subprogram execution, ~ LOCAL is a valid variable only in the MAIN program, ~ and so substitution cannot be performed, and an error occurs. END ⑤ GOTO was executed in the ON-DO statement during subtask execution.
Remedy	① Check the robot system specifications to see if it includes the conveyor synchronization option. In systems that do not include the conveyor synchronization option, remove the conveyor commands from the robot program. ② Remove the operation commands in the subtask processing of the robot program. ③ Remove the task commands in the ON-DO statement. ④ Change the auto variable in the ON-DO statement command to a global variable.
Remarks	

2-034 Ethernet connection error

Description	A PRINT/INPUT command was executed before establishing an Ethernet connection.
Cause	① The Ethernet cable is not connected. ② The parameter settings are incorrect.
Remedy	① Check the Ethernet cable connection, and turn the power off and then on again. ② Check the parameter settings.
Remarks	For details about Ethernet connection, refer to the Communication manual "Ethernet Specifications and Settings".

2-035 Config data error

Description	An invalid value was set in CONFIG.
Cause	An invalid value was set in CONFIG.
Remedy	Correct the program.
Remarks	

2-036 Vector data error

Description	An error occurred in the vector calculation processing.
Cause	① An attempt was made to convert the POINT data outside the movement range to JOINT. ② The vector data value is outside the effective range.
Remedy	Correct the program.
Remarks	

2-037 Strings too long

Description	A character string that exceeded the character string type size was entered.
Cause	An attempt was made to enter a character string that exceeded the character string type size.
Remedy	Correct the program.
Remarks	

2-039 PASS command error

Description	The PASS command cannot be executed.
Cause	ENABLE PASS was specified during ENABLE SMOOTH.
Remedy	Check that ENABLE PASS was not specified during ENABLE SMOOTH in the program.
Remarks	

2-040 Invalid SMOOTH command

Description	The SMOOTH command cannot be executed.
Cause	ENABLE SMOOTH was specified during ENABLE PASS.
Remedy	Check that ENABLE SMOOTH was not specified during ENABLE PASS in the program.
Remarks	

2-041 Invalid variable

Description	This indicates that a value specified as an argument cannot be used.
Cause	① TAN: An integer multiple of 90° was set as the argument. ② ASIN, ACOS: A value smaller than -1 or larger than 1 was set as the argument. ③ LN, LOG10, SQRT: A value of less than 0 was set as the argument. ④ DIN, BCDIN, DOUT, BCDOUT, or PULOUT: A nonexistent line number was specified. ⑤ MOVEI or MOVEA: There are four control axes, but a fifth axis was specified. ⑥ DIM: An index exceeding the maximum value of the value defined by the array command was specified. ⑦ A value other than conveyor 1 and 2 was specified in the conveyor command.
Remedy	Change the arguments for the relevant items ① to ⑦.
Remarks	

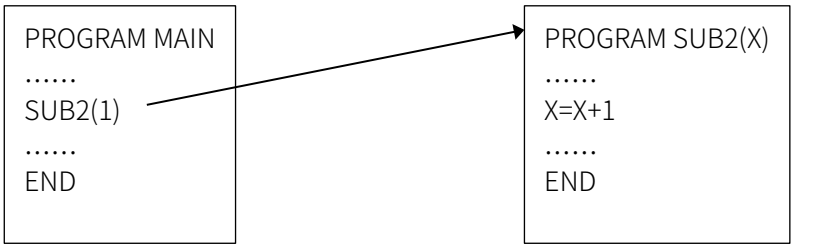
2-042 Data type error

Description	There is an error in the variable type.
Cause	① An undefined variable was used in the conditional IF statement. ② An undefined variable was used in the conditional statement in the WAIT command.
Remedy	①②Check if an undefined variable was used in a conditional statement. (IF A==1 THEN A value should have been entered in A here.)
Remarks	

2-043 Invalid ON command

Description	Specified ON conditions were erased.
Cause	Invalid processing occurred in the internal calculation of the ON condition command. There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

2-045 Write to const

Description	A program that writes a value to a constant area was executed.
Cause	An attempt was made to change the value of a transferred constant in a subprogram where a constant is used in the argument (LEFTY, P1, 1, 100, etc.). (Example)  A value cannot be substituted because X is the constant "1".
Remedy	Correct the program.
Remarks	

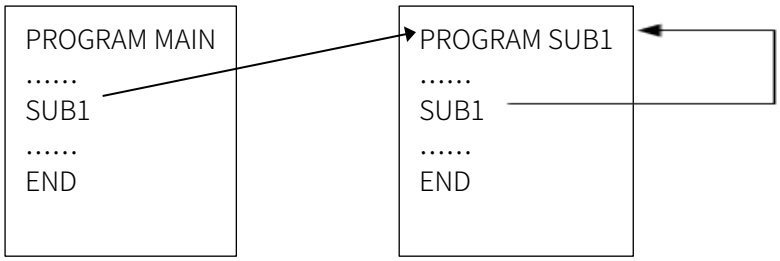
2-046 Invalid channel

Description	The specified communication channel is not valid.
Cause	① The specified communication port is not available for use. ② There is an error in the communication parameter settings. ③ A buffer overflow occurred in communication. ④ An error occurred in communication due to a broken wire in a cable or poor contact by the connector. ⑤ The main module is faulty.
Remedy	① Check if the target port is specified by the CHANNEL command. ②③ Check the settings in the Parameter Manual: ETHERNET.PAR and the parameters in the other communicating device. ④ Check if there are any broken wires in the cables. ⑤ Replace the controller.
Remarks	

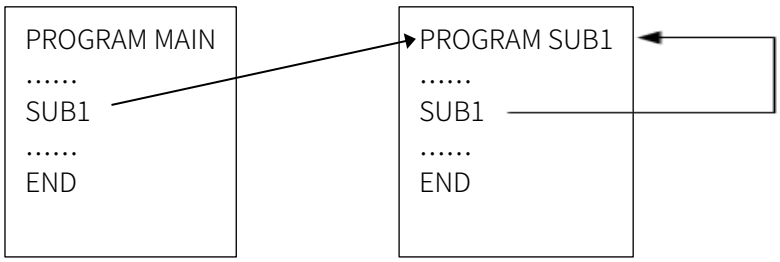
2-047 Task limits over

Description	The program tried to generate tasks that exceeded the maximum value that was set by the MAXTASK command.
Cause	① The value set by MAXTASK (number of tasks that can exist at the same time) is too small. ② Too many tasks were generated.
Remedy	① Increase the value for MAXTASK. ② Reduce the number of tasks.
Remarks	For details on the multitask processing, refer to the Robot Language Manual.

2-049 Symbol overflow

Description	The area required for internal calculation processing during SCOL program execution was not large enough.
Cause	① The area required for the symbol process used in the program was not enough. ② A recursive call is occurring in the SCOL program. 
Remedy	① Use subprograms to reduce the number of symbols in the program. Provide separate areas for the main program and the subprograms. ② Correct the program that is issuing a recursive call.
Remarks	

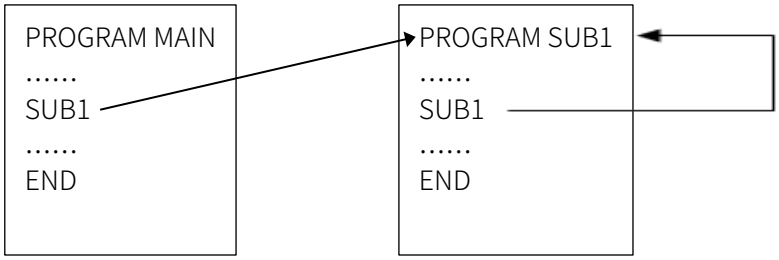
2-050 Stack overflow

Description	The area required for internal calculation processing during SCOL program execution was not large enough.
Cause	① A recursive call is occurring in the SCOL program.  ② There are too many nesting structures* of subprograms in the SCOL program.
Remedy	① Correct the program that is issuing a recursive call. ② Revise the SCOL program by reducing the number of nesting structures* of subprograms in the SCOL program.
Remarks	* A nested structure refers to calling of a separate subprogram during execution of a subprogram. When a subprogram is called, the nesting level increases by one, and when a subprogram is completed, the nesting level decreases by one.

2-051 ON-DO overflow

Description	The maximum number of ON-DO commands (condition monitor commands) that can be registered (executed) has been exceeded.
Cause	A maximum of 10 ON condition commands can be executed simultaneously. An alarm occurs if more than 10 commands were executed.
Remedy	Reduce the number of ON-DO commands.
Remarks	

2-054 Text overflow

Description	The area required for internal calculation processing during SCOL program execution was not large enough.
Cause	① The area required for processing of character strings used in the program was not large enough. ② A recursive call is occurring in the SCOL program. 
Remedy	① Use subprograms to reduce the number of symbols in the program. Provide separate areas for the main program and the subprograms. ② Correct the program that is issuing a recursive call.
Remarks	

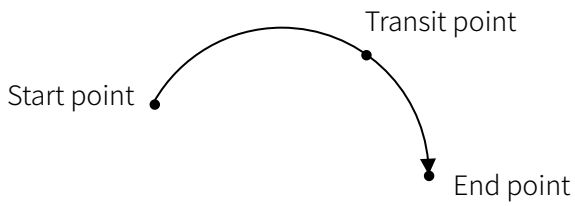
2-065 Motion command error

Description	A path operation could not be calculated in the path generation processing.
Cause	There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version, conditions when the error occurred including the line number, program where the error occurred, and all files in the controller.
Remarks	

2-066 Coordinate conversion error

Description	A conflict was found in the coordinate conversion results, and thus the robot operation cannot be performed.
Cause	There may be a fault in the system.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version, conditions when the error occurred including the line number, program where the error occurred, and all files in the controller.
Remarks	

2-067 MOVEC error

Description	A point that cannot be generated as an arc was specified in the MOVEC command.
Cause	① The coordinates for the same point are specified as the start point, transit point, and end point of the MOVEC command. ② The coordinates for the start point, transit point, and end point of the MOVEC command are on the same line.
Remedy	Specify the start point, transit point, and end point of the MOVEC command so that they are all different and are not on the same line. 
Remarks	

2-068 PAYLOAD error

Description	Execution of the PAYLOAD command failed.
Cause	The argument in the PAYLOAD command is invalid.
Remedy	Check the argument in the PAYLOAD command of the SCOL program.
Remarks	PAYLOAD command format PAYLOAD = {<Weight>, <Offset of center of gravity>}

2-069 CONFIG error

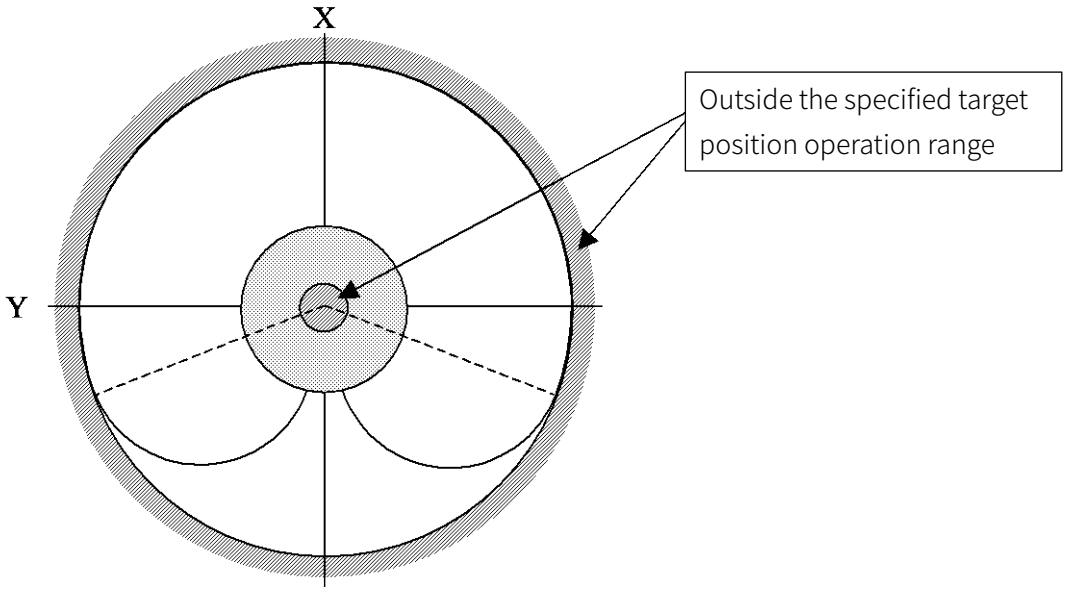
Description	An error was found in the specified posture in the CP movement command[1)] (MOVES command, MOVEC command).
Cause	① The postures of the start point and end point in the MOVES command are different. ② The postures of the start point, transit point, and end point in the MOVEC command are different. ③ This alarm occurs when program execution is stopped during interpolation operation, then manual guidance or human power is used to change the robot posture, and the program is executed again.
Remedy	Check that the start point, transit point (circle), and end point of the interpolation operation have the same teach configuration (LEFTY, RIGHTY).
Remarks	

① Command that continuously interpolates between the predetermined sites to enable tracking of the robot.
The MOVES command performs linear interpolation, and the MOVEC command performs arc interpolation.

2-071 MOVE CV calculation error

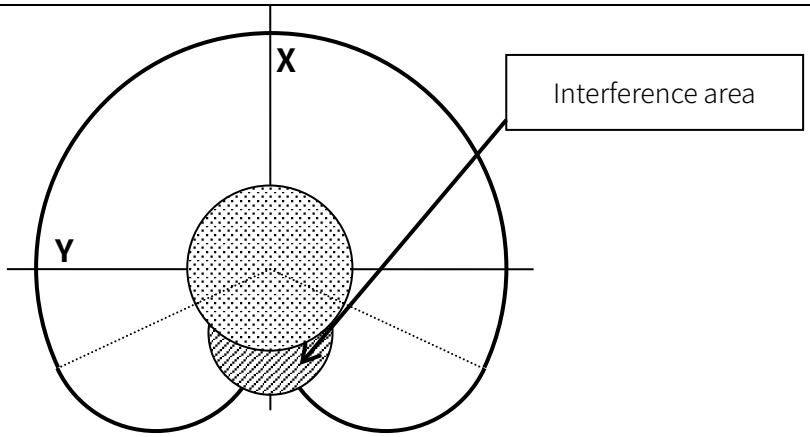
Description	The correct orbit could not be calculated with the MOVE CV command.
Cause	① It contains teaching points where the C-axis angle changes significantly. ② The same teaching points are continuous.
Remedy	Correct the teaching points.
Remarks	

2-079 Point out of limit

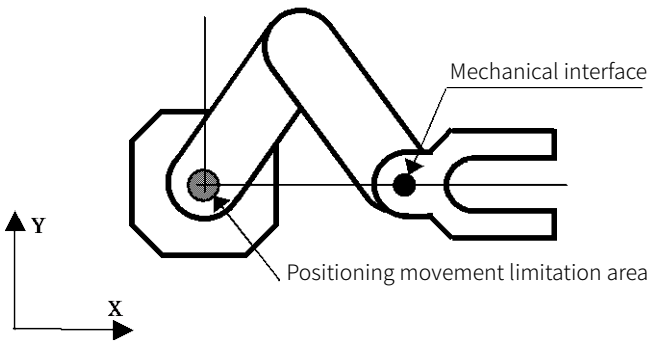
Description	Operation cannot be performed because a position where movement is not possible was specified in the program operation command.
Cause	① A position outside the arm length was specified. (Point that cannot be reached even by extending the arm) ② A position inside the Axis 1 length – Axis 2 length was specified. (Point that cannot be reached even by folding over the arm).
Remedy	Revise the specified positions (teaching points).
Remarks	The area outside the target position operation range of the SCARA robot is shown in the figure below. 

2-081 Destination interfere

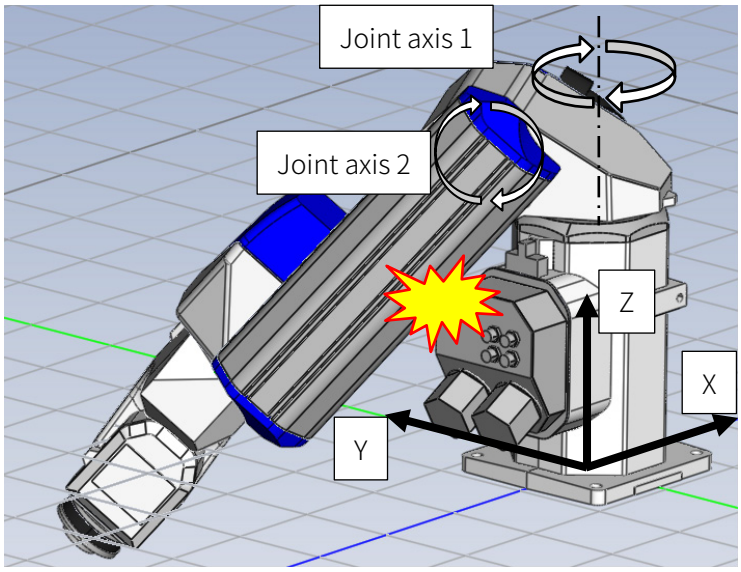
Description	Operation cannot be performed because a position within the interference area was specified in a program operation command.
Cause	A position within the interference area was specified in the program.
Remedy	Revise the specified positions (teaching point data).
Remarks	An example of the interference area of a SCARA robot is shown in the figure below. The interference area is set for preventing interference between the robot and the robot body.



2-082 Outside limitation area

Description	An invalid position was specified as the target position.
Cause	An invalid position was specified as the target position.
Remedy	Position the robot program (teach points), and correct so that they are not within the positioning limitation area.
Remarks	

2-083 Movement limitation area

Description	A position in the movement limitation area was specified as the target operation position in the program operation command.
Cause	The target operation position of the program was specified in an area where interference may occur due to the combined operation of multiple joint axes. This alarm is used to prevent interference with the robot itself (base, arm, cables, etc.) due to combined movements of different joint axes. The movement limitation areas of the joint axes are preset in each robot.  Example of interference between the arm and the body due to combined movement of axes No. 1 and No. 2 If the target operation position is within the movement limitation area, the robot decelerates to stop before the operation command is executed.
Remedy	Review the target operation position, and correct it to a position where interference with the arm can be avoided. Use jog operation to move in the opposite direction from the movement direction just before the alarm occurred. (The alarm is automatically cleared.)
Remarks	This alarm occurs in vertically articulated robots only.

2-084 Outside limitation area

Description	The target operation position is outside the limitation area.
Cause	The target operation position is outside the limitation area set in [A001] of AREA.PAR.
Remedy	Revise the specified positions (teaching point data).
Remarks	

2-085 No entry area

Description	The target operation position is in the prohibited area.
Cause	The target operation position is in the prohibited area set in [A002] of AREA.PAR.
Remedy	Revise the specified positions (teaching point data).
Remarks	

2-089 MOVEJ parameter error

Description	The argument of the MOVEJ command of the robot program is invalid.
Cause	The argument of the MOVEJ command of the robot program is invalid.
Remedy	Correct the argument of the MOVEJ command in the program.
Remarks	MOVEJ command format MOVEJ <position> {<retract operation travel distance>, <single-axis operation travel distance 1>, <single-axis operation travel distance 2>}

2-097 C Axis MOVE error

Description	A large C-axis movement command was performed during the SMOOTH motion.
Cause	SMOOTH motion is impossible because there is a C-axis MOVE command specifying a large motion during SMOOTH motion (ENABLE SMOOTH).
Remedy	Check the target position (teach point) in SMOOTH motion, and correct the C-axis position.
Remarks	

2-098 T Axis impossible

Description	A T-axis movement command was performed during the SMOOTH motion.
Cause	The T-axis cannot be moved during SMOOTH motion (ENABLE SMOOTH).
Remedy	Check the target position (teach point) in SMOOTH motion, and correct the T-axis position.
Remarks	

2-129 User alarm
2-130 User alarm
2-131 User alarm
2-132 User alarm
2-133 User alarm
2-134 User alarm
2-135 User alarm
2-136 User alarm

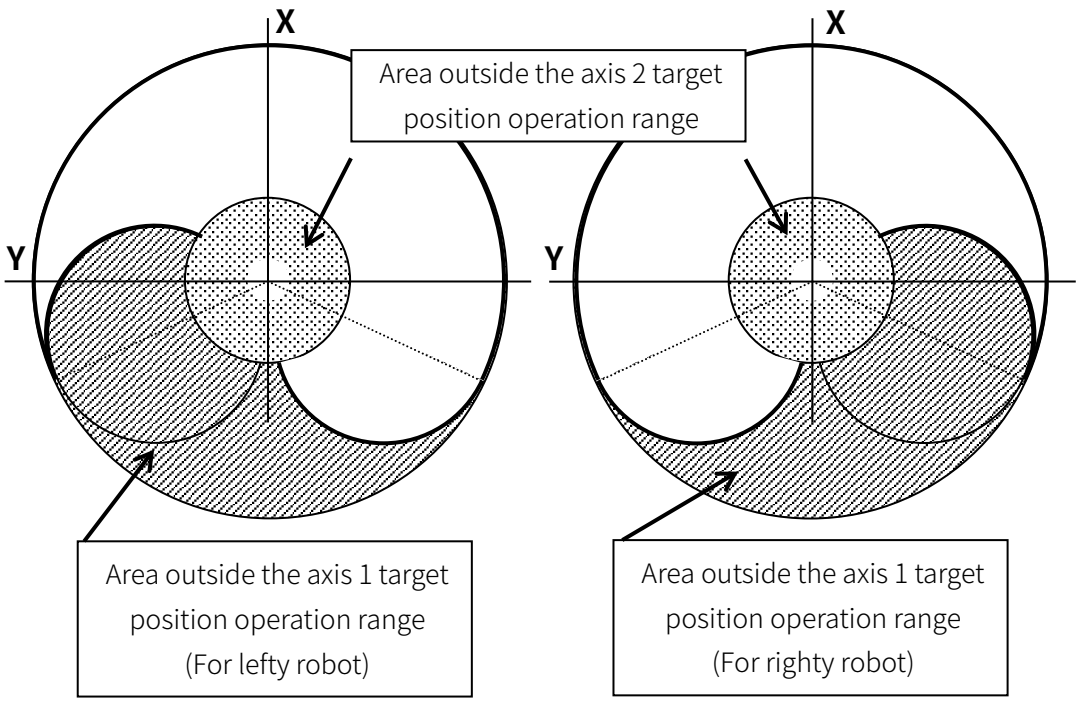
Description	A level 2 user alarm occurred.
Cause	A level 2 user alarm occurred.
Remedy	Remove the alarm conditions that were set by the customer.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual "[U026] USER ALARM".

2-145 IP1 status alarm
2-146 IP2 status alarm
2-147 IP3 status alarm
2-148 IP4 status alarm
2-149 IP5 status alarm
2-150 IP6 status alarm
2-151 IP7 status alarm
2-152 IP8 status alarm

Description	The Ethernet connection was disconnected during running of the program.
Cause	The Ethernet connection was disconnected.
Remedy	Check the connection between the robot controller and the device connected via Ethernet. Check that the Ethernet cable is not damaged.
Remarks	The occurrence and level for this alarm can be set in the ETHERNET.PAR file.

2-401	Axis1 out of Limit
2-402	Axis2 out of Limit
2-403	Axis3 out of Limit
2-404	Axis4 out of Limit
2-405	Axis5 out of Limit
2-406	Axis6 out of limit
2-407	Axis7 out of Limit
2-408	Axis8 out of Limit

Description	Movement cannot be performed because a position outside the software limit was specified in the program movement command.
Cause	① A position outside the software limit is specified. ② The software limit settings are invalid.
Remedy	① Revise the specified positions (teaching points). ② Revise the software limit settings.
Remarks	An example that occurs outside the axis No. 1 and axis No. 2 target position movement range of the SCARA robot is shown in the figure below.



2-411 Axis1 motor overload(15m)
2-412 Axis2 motor overload(15m)
2-413 Axis3 motor overload(15m)
2-414 Axis4 motor overload(15m)
2-415 Axis5 motor overload(15m)
2-416 Axis6 motor overload(15m)
2-417 Axis7 motor overload(15m)
2-418 Axis8 motor overload(15m)

Description	The 15-minute running average of the motor load exceeded a predetermined value.
Cause	A load exceeding a specified value was applied to the motor.
Remedy	Reduce the load applied to the motor. • Use the PAYLOAD command to set an appropriate load value. • Check that the weight of the workpiece + hand does not exceed the value in the specifications.
Remarks	

2-421 Axis1 motor overload(1m)
2-422 Axis2 motor overload(1m)
2-423 Axis3 motor overload(1m)
2-424 Axis4 motor overload(1m)
2-425 Axis5 motor overload(1m)
2-426 Axis6 motor overload(1m)
2-427 Axis7 motor overload(1m)
2-428 Axis8 motor overload(1m)

Description	The 1-minute running average of the motor load exceeded a predetermined value.
Cause	A load exceeding a specified value was applied to the motor.
Remedy	Reduce the load applied to the motor. • Use the PAYLOAD command to set an appropriate load value. • Check that the weight of the workpiece + hand does not exceed the value in the specifications.
Remarks	

2-471 Axis1 motor overload(2h)
2-472 Axis2 motor overload(2h)
2-473 Axis3 motor overload(2h)
2-474 Axis4 motor overload(2h)
2-475 Axis5 motor overload(2h)
2-476 Axis6 motor overload(2h)
2-477 Axis7 motor overload(2h)
2-478 Axis8 motor overload(2h)

Description	The 2-hour running average of the motor load exceeded a predetermined value.
Cause	A load exceeding a specified value was applied to the motor.
Remedy	Reduce the load applied to the motor. • Use the PAYLOAD command to set an appropriate load value. • Check that the weight of the workpiece + hand does not exceed the value in the specifications.
Remarks	

2-561 Axis1 overheat
2-562 Axis2 overheat
2-563 Axis3 overheat
2-564 Axis4 overheat
2-565 Axis5 overheat
2-566 Axis6 overheat
2-567 Axis7 overheat
2-568 Axis8 overheat

Description	The specified motor temperature was exceeded.
Cause	The motor has overheated.
Remedy	Reduce the load applied to the motor. • Use the PAYLOAD command to set an appropriate load value. • Check that the weight of the workpiece + hand does not exceed the value in the specifications.
Remarks	

Level 1 Alarms

This chapter describes the level 1 alarms that occur in the TS5000, their causes, and corrective actions. When a level 1 alarm is detected, the robot displays the alarm message only.

1-003 M-TO CONFIG err

Description	A configuration error occurred when a teach point was moved.
Cause	When teach point movement was performed by linear interpolating operation, the configuration at completion of teach point movement and the teach point configuration differed.
Remedy	Change the teach point movement from linear interpolating operation (MOVES) to MOVE operation.
Remarks	If the targeted teach point configuration contains FREE, no alarm occurs regardless of the configuration after completion of teach point movement.

1-009 Unable to RCYCLE

Description	The cycle reset was invalid.
Cause	An attempt was made to stop operation during execution of the DO statement in a multi-task program or ON-DO command and to use this function.
Remedy	The cycle reset command cannot be used during execution of a DO statement in a multi-task program or ON-DO command. Either change the location where the cycle reset command is used, or correct the SCOL program.
Remarks	

1-015 Parameter error

Description	A parameters error was detected.
Cause	An unrecognizable key was found in the parameter file. The value that was set in the parameter file is invalid.
Remedy	Check the setting value for the line number where the parameter error occurred. Delete the key for the line number where the parameter error occurred.
Remarks	

1-017 PASS command error

Description	The ENABLE PASS is invalid.
Cause	An ENABLE PASS command was specified during an ENABLE SMOOTH command.
Remedy	Correct the SCOL program.
Remarks	

1-018 Invalid SMOOTH command

Description	The ENABLE SMOOTH is invalid.
Cause	An ENABLE SMOOTH command was specified during an ENABLE PASS command.
Remedy	Correct the SCOL program.
Remarks	

1-025 Over deceleration

Description	Deceleration was performed at a rate that exceeds the parameter value in the operation command specified by the DISABLE SMOOTH command.
Cause	The travel for the final operation in the program during ENABLE PASS is short, and deceleration was performed at an acceleration that exceeded the parameter value.
Remedy	Revise the SCOL program (teaching points), and extend the travel distance of the movement using DISABLE SMOOTH.
Remarks	

1-026 SMOOTH impossible

Description	The SMOOTH command is disabled.
Cause	The travel distance in the SMOOTH connection program is too short, and the internal calculation cannot be performed in time.
Remedy	Correct the SCOL program (teach points), and increase the distance between teach points.
Remarks	

1-028 Already enable SW ON

Description	The ON status of the enable switch on the teach pendant was detected when the power was turned on.
Cause	① The enable switch on the teach pendant was already pressed when the power was turned on. □ The enable switch of the teach pendant has failed.
Remedy	① Set the enable switch to OFF, and then turn the power off and then on again. ② Replace the teach pendant.
Remarks	If the power is turned on when the enable switch was already pressed, the controller assumes that the enable switch remains pressed due to a malfunction.

1-037 PLC user alarm
1-038 PLC user alarm
1-039 PLC user alarm
1-040 PLC user alarm
1-041 PLC user alarm
1-042 PLC user alarm
1-043 PLC user alarm
1-044 PLC user alarm

Description	A level 1 user alarm has occurred in the internal PLC.
Cause	A level 1 user alarm has occurred in the internal PLC. The message can be defined by the user parameters.
Remedy	The user must remove the alarm conditions that were set in the internal PLC.
Remarks	

1-081 Hand 2 module detection

Description	The hand 2 module option is not enabled.
Cause	The hand 2 module option is not enabled.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

1-082 Option1 I/O module detection

Description	The option1 I/O module option is not enabled.
Cause	The option1 I/O module option is not enabled.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

1-083 Option3 conveyor synchronizing cable detection

Description	Option3 conveyor option is not enabled.
Cause	Option3 conveyor option is not enabled.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

1-084 Option4 AD module detection

Description	Option4 AD module option is not enabled.
Cause	Option4 AD module option is not enabled.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

1-085 Hand 1 module detection

Description	The hand 1 module option is not enabled.
Cause	The hand 1 module option is not enabled.
Remedy	Contact SHIBAURA MACHINE Customer Service, and provide details about the system software version and all files in the controller, including the robot program.
Remarks	

1-097 Initial IP address mode

Description	The initial IP address mode has been selected.
Cause	The initial IP address mode has been selected.
Remedy	When the selected mode is correct, no action is required.
Remarks	The IP address will be set to "192.168.0.124".

1-099 FRAM0 erase mode

Description	The FRAM0 erase mode has been selected.
Cause	The FRAM0 erase mode has been selected.
Remedy	When the selected mode is correct, no action is required.
Remarks	The FRAM0 space will be erased. FRAM0 stores maintenance information, master mode, and information on selected files.

1-101 Parameter file deletion mode

Description	The parameter file deletion mode has been selected.
Cause	The parameter file deletion mode has been selected.
Remedy	When the selected mode is correct, no action is required.
Remarks	

1-108 Memory erase mode

Description	The memory erase mode has been selected.
Cause	The memory erase mode has been selected.
Remedy	When the selected mode is correct, no action is required.
Remarks	The FRAM0 space and FRAM1 space will be erased. FRAM0 stores maintenance information, master mode, and information on selected files. FRAM1 stores memory variable information.

1-129 User alarm
1-130 User alarm
1-131 User alarm
1-132 User alarm
1-133 User alarm
1-134 User alarm
1-135 User alarm
1-136 User alarm

Description	A level 1 user alarm occurred.
Cause	A level 1 user alarm occurred.
Remedy	Remove the alarm conditions that were set by the customer.
Remarks	The message can be defined based on the parameter settings. For details, refer to the Parameter Manual "[U026] USER ALARM".

1-145 Main module battery error

Description	The battery that updates the calendar date and time has low power.
Cause	The battery that updates the calendar date and time has low power.
Remedy	Change the battery by referring to the Controller Manual.
Remarks	Leaving the main module battery alarm on will initialize the calendar date and time. Change battery as soon as possible.

1-147 COM buffer over

Description	The receiver buffer area in non-protocol communication was not found.
Cause	The amount of input data has exceeded the receiver buffer (256 bytes).
Remedy	DC3 and other flow control is not performed in non-protocol communication (COM*). Read all data received from external equipment by the INPUT command. When this alarm is reset, the receiver buffer is also cleared.
Remarks	This alarm does not occur in HOST command communication.

1-149 Extern SELECT error

Description	An error occurred in the external file selection processing.
Cause	① There is no specified file. ② There is an error in the file selection signal line setting. • External selection file (EXTRNSEL.PAR). ③ A file was selected by an external signal during program execution (during RUN).
Remedy	① Check if the specified file exists. ② Check if there is an error in the file selection signal line setting (EXTRNSEL.PAR [X000] SPECIFY SIGNAL FOR EXTSELECT) of the user parameter file. Check if there is an error in the external selection file setting (EXTRNSEL.PAR). Is the file selection number by the file selection signal line incorrect? ③ Use an external signal to select a file while the program is stopped.
Remarks	For details, refer to the Interface Manual.

1-150 Data block duplication error

Description	A data section is contained in the program file.
Cause	A data section is found in the program file even though common data section functions are being used.
Remedy	Delete the data section from the program file.
Remarks	

1-154 Data block format error

Description	A data format error occurred during startup of the data editor.
Cause	A data format error occurred during startup of the data editor.
Remedy	The data line where the error occurred can be deleted after reading. After confirmation, re-enter the correct data.
Remarks	

1-158 Short of working area

Description	Program editing can no longer be performed by the teach pendant because there is not enough available space in the controller.
Cause	Workspace for program editing cannot be provided because there is not enough available space in the controller.
Remedy	Move or delete any unneeded files in the controller.
Remarks	

1-160 Compile error

Description	The program could not be compiled.
Cause	A syntax error occurred in the file selection processing.
Remedy	Check the error description on the alarm display screen. After checking, revise the problem location in the program.
Remarks	For more information about the detailed display of the compile errors, refer to the instruction manual of the respective device and software.

1-164 PLC backup data error

Description	A problem occurred in the PLC ladder program.
Cause	① The ladder program is corrupted. ② The ladder program is not saved.
Remedy	Resend the ladder program using TC-WORX.
Remarks	

1-166 PLC remote unit error

Description	An extended I/O module error occurred.
Cause	① Extended I/O module connection settings do not match with those of user parameters. ② The cables are not connected correctly, or they have a broken wire. ③ Extended I/O module connection settings do not match with those of user parameters.
Remedy	① If the user parameter settings are incorrect, set the correct user parameters. ② Check the connection cables, and replace if anything is found wrong. ③ Please contact SHIBAURA MACHINE Customer Service if the I/O module is defective.
Remarks	

1-169 PLC undefine label

Description	The label that is being used in the PLC ladder program is not defined.
Cause	The label that is being used in the PLC ladder program is not defined.
Remedy	Revise the ladder program using TC-WORX.
Remarks	

1-170 PLC invalid command

Description	The command that is being used in the PLC ladder program is not defined.
Cause	The command that is being used in the PLC ladder program is not defined.
Remedy	Revise the ladder program using TC-WORX.
Remarks	

1-173 PLC overlap label

Description	The label that is being used in the PLC ladder program is defined twice.
Cause	The label that is being used in the PLC ladder program is defined twice.
Remedy	Revise the ladder program using TC-WORX.
Remarks	

1-194 Can not use SD card

Description	The inserted SD card cannot be used.
Cause	① The SD card is not formatted. ② The SD card is incompatible.
Remedy	Format the SD card in the FAT file system. Use an SD card compatible with the SD or SDHC standards.
Remarks	

1-201 IP1 status alarm
1-202 IP2 status alarm
1-203 IP3 status alarm
1-204 IP4 status alarm
1-205 IP5 status alarm
1-206 IP6 status alarm
1-207 IP7 status alarm
1-208 IP8 status alarm

Description	The Ethernet connection was disconnected during running of the program.
Cause	The Ethernet connection was disconnected.
Remedy	Check the connection between the robot controller and the device connected via Ethernet. Check that the Ethernet cable is not damaged.
Remarks	The occurrence and level for this alarm can be set in the ETHERNET.PAR file.

1-209 Robot serial number mismatch

Description	The combination of robots connected to the controller is incorrect.
Cause	A robot different from the factory combination was connected to the controller.
Remedy	Reconnect the controller to the correct combination of robots.
Remarks	

1-401	Axis1 Enc battery low
1-402	Axis2 Enc battery low
1-403	Axis3 Enc battery low
1-404	Axis4 Enc battery low
1-405	Axis5 Enc battery low
1-406	Axis6 Enc battery low
1-407	Axis7 Enc battery low
1-408	Axis8 Enc battery low

Description	The voltage of the battery used to store the encoder position is low.
Cause	The voltage of the battery used to store the encoder position is low.
Remedy	Change the battery by referring to the Robot Manual.
Remarks	If the encoder battery alarm is left without taking any action, the robot origin could be shifted, or other major problem could occur. Change the battery as soon as possible

1-421	Axis1 failed to read from encoder EEPROM
1-422	Axis2 failed to read from encoder EEPROM
1-423	Axis3 failed to read from encoder EEPROM
1-424	Axis4 failed to read from encoder EEPROM
1-425	Axis5 failed to read from encoder EEPROM
1-426	Axis6 failed to read from encoder EEPROM
1-427	Axis7 failed to read from encoder EEPROM
1-428	Axis8 failed to read from encoder EEPROM

Description	Failed to read the data from the encoder EEPROM.
Cause	Failed to access the encoder EEPROM.
Remedy	Restart the controller. If it occurs frequently, replace the servo motor.
Remarks	

1-431	Axis1 failed to write to encoder EEPROM
1-432	Axis2 failed to write to encoder EEPROM
1-433	Axis3 failed to write to encoder EEPROM
1-434	Axis4 failed to write to encoder EEPROM
1-435	Axis5 failed to write to encoder EEPROM
1-436	Axis6 failed to write to encoder EEPROM
1-437	Axis7 failed to write to encoder EEPROM
1-438	Axis8 failed to write to encoder EEPROM

Description	Failed to write data to encoder EEPROM.
Cause	Failed to access the encoder EEPROM.
Remedy	Re-execute the executed operation. If it occurs frequently, replace the servo motor.
Remarks	

Appendix

1. Controller Module Names

The following shows the module names of the controller.

	TS5000-MS	TS5000-SS	TS5000-EMS
Main Module	TS5K-MAIN	TS5K-MAIN	TS5K-MAIN
Servo Amplifier Module	TS5KSV-M4	TS5KSV-S4	TS5KSV-EM4
Servo Power Module	X8KP	X8KZ	X8KV
Base Module	X8KM	X8KI	X8KX
System I/O Module	TS5K-SYIO	TS5KS-SYIO	TS5KS-SYIO
Extended I/O Module	TS5K-OPIO	TS5KS-OPIO	TS5KS-OPIO
PROFIBUS Module	TS5K-PB	TS5K-PB	TS5K-PB
DeviceNet Module	TS5K-DN	TS5K-DN	TS5K-DN
EtherNet/IP Module	TS5K-IP	TS5K-IP	TS5K-IP
CC-Link Module	TS5K-CC	TS5K-CC	TS5K-CC
PROFINET Module	TS5K-PN	TS5K-PN	TS5K-PN
EtherCAT Module	TS5K-CA	TS5K-CA	TS5K-CA
AD Module	X8KD	X8KD	X8KD

2. Alarm Message Language Correspondence Table

The alarm messages vary depending on the language setting.

If the TP1000 is used, the language setting is fixed at English, and so refer to the table below.

Alarm Message Language Correspondence Table

Number	English	Japanese	Chinese	Korean
008-001	Undefined Interrupt	未定義割り込み発生	发生未定义中断	미정의 인터럽트 발생
008-004	Watchdog error	ウォッチドッグ異常	监控器错误	워치도그 에러
008-005	PLC CPU error	PLC CPU 異常	PLC CPU 异常	PLC CPU 이상
008-007	Servo soft Watchdog1 error	サーボソフトウォッチドッグ 1 異常	伺服软件监控器 1 异常	서보 소프트 워치도그 1 이상
008-008	Servo soft Watchdog2 error	サーボソフトウォッチドッグ 2 異常	伺服软件监控器 2 异常	서보 소프트 워치도그 2 이상
008-013	AC Alarm	AC 電源異常	AC 报警	AC 경고
008-014	Emergency Stop SW ON	非常停止スイッチオン	紧急停止中	비상정지 중
008-017	Enable SW OFF	イネーブルスイッチオフ	安全开关 ON	안전 스위치 ON
008-018	Servo CPU error	サーボ CPU 異常	伺服 CPU 异常	서보 CPU 이상
008-019	Already Enable SW ON	起動時イネーブルスイッチオン	启用开关 ON 状态	이네이블 스위치 ON 상태
008-020	Servo amp module model mismatch	サーボアンプモジュール型式不一致	伺服板型号不匹配	서보 보드 모델 불일치
008-023	PN Voltage Error	PN 電圧異常検出	PN 电压误差	PN 전압 오류
008-024	RGST Error	RGST エラー	RGST 异常	RGST 이상
008-025	Servo PL WDT Error	PL ウォッチドッグ異常	PL 监控器异常	PL 워치도그 이상
008-027	Slow Charge error	スローチャージ異常	慢速充电错误	완충전 에러
008-030	Servo Power Low Voltage	サーボ電源過小電圧	伺服电源电压过低	서보 전원 과소 전압
008-032	Servo Power Over Voltage	サーボ電源過大電圧	伺服电源电压过大	서보 전원 과대 전압
008-033	Distribution error	分配処理異常	分配处理运算错误	분배 처리 연산 에러
008-034	Unable to Trans	目標位置座標変換不可	不可转换分配目标位置坐标	분배 목표 위치 좌표 변환 불가
008-035	Point Invalid of Range	目標位置座標範囲外	分配目标位置动作偏离范围错误	분배 목표 위치 동작범위 외 에러
008-04X	AxisX Exceed Speed	第 X 関節軸 速度指令値過大	第 X 轴 速度指令值过大	제 X 축 속도 지령값 과대
008-065	Main CPU Peripheral failure	Main CPU Peripheral 失敗	Main CPU Peripheral 失败	Main CPU Peripheral 실패
008-066	QSPI ROM Initialize failure	QSPI ROM 初期化失敗	QSPI ROM 初始化失败	QSPI ROM 초기화 실패
008-067	Io-EX Initialize failure	Io-EX 初期化失敗	Io-EX 初始化失败	Io-EX 초기화 실패
008-068	RTC Initialize failure	RTC 初期化失敗	RTC 初始化失败	RTC 초기화 실패
008-069	ADC Device Initialize failure	ADC 初期化失敗	ADC 初始化失败	ADC 초기화 실패
008-070	LAN Initialize failure	LAN 初期化失敗	LAN 初始化失败	LAN 초기화 실패
008-071	LAN1 PHY Address illegal	LAN1 PHY アドレス異常	LAN1 PHY 地址异常	LAN1 PHY 어드레스 이상
008-072	LAN2 PHY Address illegal	LAN2 PHY アドレス異常	LAN2 PHY 地址异常	LAN2 PHY 어드레스 이상
008-073	LAN1 Config failure	LAN1 設定失敗	LAN1 设定失败	LAN1 설정 실패
008-074	LAN2 Config failure	LAN2 設定失敗	LAN2 设定失败	LAN2 설정 실패
008-075	Communication Start failure	通信開始失敗	通信开始失败	통신 개시 실패
008-076	Io-EX0 Communication failure	Io-EX0 通信失敗	Io-EX0 通信失败	Io-EX0 통신 실패
008-077	Io-EX1 Communication failure	Io-EX1 通信失敗	Io-EX1 通信失败	Io-EX1 통신 실패
008-078	Io-EX2 Communication failure	Io-EX2 通信失敗	Io-EX2 通信失败	Io-EX2 통신 실패
008-079	RTC Communication failure	RTC 通信失敗	RTC 通信失败	RTC 통신 실패
008-080	ADC Communication failure	ADC 通信失敗	ADC 通信失败	ADC 통신 실패
008-081	Servo Task1 WDT Error	サーボタスク 1WDT 異常	伺服任务 1WDT 异常	서보 태스크 1WDT 이상
008-082	Servo Task2 WDT Error	サーボタスク 2WDT 異常	伺服任务 2WDT 异常	서보 태스크 2WDT 이상
008-083	Servo Main WDT1 Error	サーボ・メイン間 WDT1 異常	伺服·主控间 WDT1 异常	서보・메인 간 WDT1 이상
008-084	Servo Main WDT2 Error	サーボ・メイン間 WDT2 異常	伺服·主控间 WDT2 异常	서보・메인 간 WDT2 이상
008-113	Robot Parameter Error	ロボットパラメータ異常	机械手参数异常	로봇 파라미터 이상
008-114	Servo Parameter Error	サーボパラメータ異常	伺服参数异常	서보 파라미터 이상
008-115	FRAM0 Area CRC error	FRAM0 空間 CRC 異常	FRAM0 区 CRC 错误	FRAM0 영역 CRC 오류
008-116	FRAM1 Area CRC error	FRAM1 空間 CRC 異常	FRAM1 区 CRC 错误	FRAM1 영역 CRC 오류
008-117	FRAM0 Area Write Fail	FRAM0 空間書き込み失敗		
008-129	User Alarm	ユーザアラーム	用户报警	사용자 경고

Number	English	Japanese	Chinese	Korean
008-130	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-131	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-132	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-133	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-134	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-135	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-136	User Alarm	ユーザアラーム	用户报警	사용자 경고
008-241	Controller Type error	コントローラ型名異常	控制器型号名称异常	컨트롤러 형명 이상
008-242	CUNET is failure	CUNET 故障	CUNET 故障	CUNET 고장
008-243	System I/O Module failure	システム I/O モジュール故障	系统 I/O 基板故障	시스템 I/O 기판 고장
008-244	Hand1 Module failure	ハンド 1 モジュール故障	Hand1 基板故障	Hand1 기판 고장
008-245	Hand2 Module failure	ハンド 2 モジュール故障	Hand2 基板故障	Hand2 기판 고장
008-246	Option1 I/O Module failure	Option1 I/O モジュール故障	Option1 I/O 基板故障	Option1 I/O 기판 고장
008-248	Option3 Conveyor Cable failure	Option3 コンベア同期用ケーブル故障		
008-250	Option4 Type error	Option4 型名異常	Option4 型号名称异常	Option4 형명 이상
008-251	Option4 AD Module failure	Option4 AD モジュール故障	Option4 AD 基板故障	Option4 AD 기판 고장
008-252	Option4 I/O Module failure	Option4 I/O モジュール故障	Option4 I/O 基板故障	Option4 I/O 기판 고장
008-258	Backup Memory error	バックアップメモリ異常	备份存储器错误	백업 메모리 에러
008-269	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
008-270	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
008-271	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
008-272	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
008-273	I/O Fuse Broken	I/O フューズ破損	I/O 保险丝错误	I/O 퓨즈 에러
008-274	Enable SW error	イネーブルスイッチ異常	启用开关错误	이네이블 스위치 이상
008-275	EMG SW error	非常停止異常	紧急停止开关错误	비상정지 스위치 에러
008-277	PLC STOP	PLC ストップ	PLC 停止报警	PLC 정지 에러
008-278	Ext DC24V Regulator error	外部 24V 電源異常	外部 24V 电源异常	외부 24V 전원 이상
008-279	Ext Resistor Overheat	外部抵抗過熱	外部再生电阻过热	외부 회생 저항 과열
008-283	Illegal master mode key	マスターモードキー操作異常	主站模式键操作错误	마스터 모드 키 조작 이상
008-284	Controller Mode SW error	コントローラモードスイッチ異常	控制器模式开关异常	컨트롤러 모드 스위치 이상
008-285	Master mode mismatch	マスターモード不一致	主站模式切换时发生不一致	마스터 모드 불일치
008-354	Fieldbus Offline	フィールドバスオフライン	现场总线离线	필드버스 오프라인
008-355	Fieldbus Module error	フィールドバスモジュール異常	现场总线基板异常	필드버스 기판 이상
008-361	IP1 status alarm	IP1 ステータス異常	IP1 状态报警	IP1 상태 경고
008-362	IP2 status alarm	IP2 ステータス異常	IP2 状态报警	IP2 상태 경고
008-363	IP3 status alarm	IP3 ステータス異常	IP3 状态报警	IP3 상태 경고
008-364	IP4 status alarm	IP4 ステータス異常	IP4 状态报警	IP4 상태 경고
008-365	IP5 status alarm	IP5 ステータス異常	IP5 状态报警	IP5 상태 경고
008-366	IP6 status alarm	IP6 ステータス異常	IP6 状态报警	IP6 상태 경고
008-367	IP7 status alarm	IP7 ステータス異常	IP7 状态报警	IP7 상태 경고
008-368	IP8 status alarm	IP8 ステータス異常	IP8 状态报警	IP8 상태 경고
008-41X	AxisX Failed to Read Origin Information	第 X 関節軸 原点情報読込失敗	第 X 轴 原产地信息读取失败	제 X 축 원산지 정보 읽기 실패
008-42X	AxisX Position error	第 X 関節軸 位置誤差過大異常	第 X 轴 位置误差过大异常	제 X 축 위치 오차 과대
008-45X	AxisX Motor Locked	第 X 関節軸 モータロック	第 X 轴 马达锁定	제 X 축 모터 록
008-46X	AxisX Motor speed over	第 X 関節軸 モータオーバースピード	第 X 轴 过速	제 X 축 오버 스피드
008-48X	AxisX IPM Error	第 X 関節軸 IPM 異常	第 X 轴 IPM 异常	제 X 축 앰프 에러
008-49X	AxisX Electric Thermal	第 X 関節軸 電気サーマル異常	第 X 轴 电子热动继电器检测	제 X 축 전자 서멀 검출
008-51X	AxisX Current Feedback	第 X 関節軸 電流フィードバック異常	第 X 轴 电流反馈错误	제 X 축 전류 피드백 에러
008-52X	AxisX Servo Parameter	第 X 関節軸 サーボパラメータ異常	第 X 轴 执行参数计算错误	제 X 축 실행 파라미터 산출 에러
008-53X	AxisX Over current	第 X 関節軸 過電流検知	第 X 轴 过电流	제 X 축 과전류 검지
008-54X	AxisX Amp rotary switch error	第 X 関節軸 ロータリスイッチ設定異常	第 X 轴 放大器旋转开关错误	제 X 축 앰프 로터리 스위치 에러
008-56X	AxisX Communication error	第 X 関節軸 定常通信異常	第 X 轴 码器断线错误	제 X 축 인코더 단선 에러
008-57X	AxisX Encoder type error	第 X 関節軸 エンコーダ形式異常	第 X 轴 编码器识别错误	제 X 축 인코더 식별 에러

Number	English	Japanese	Chinese	Korean
008-58X	AxisX Encoder over speed	第 X 関節軸 エンコーダスピード過大	第 X 轴 编码器过速	제 X 축 인코더 오버 스피드
008-59X	AxisX Encoder single count error	第 X 関節軸 シングルターンカウント異常	第 X 轴 轴编码器单圈计数错误	제 X 축 인코더 싱글 턴 카운트 에러
008-60X	AxisX Encoder multi count error	第 X 関節軸 マルチターンカウント異常	第 X 轴 编码器多圈计数错误	제 X 축 인코더 멀티 턴 카운트 에러
008-61X	AxisX Encoder battery empty	第 X 関節軸 エンコーダバッテリ枯渇	第 X 轴 编码器电池电量耗尽	제 X 축 인코더 배터리 엠티
008-72X	AxisX Abnormal U phase Current Offset Value	第 X 関節軸 U 相電流オフセット値異常検知	第 X 轴 U 相电流偏移值异常检测	제 X 축 U 상 전류의 오프셋 값 이상 검출
008-73X	AxisX Abnormal V phase Current Offset Value	第 X 関節軸 V 相電流オフセット値異常検知	第 X 轴 V 相电流偏移值异常检测	제 X 축 V 상 전류의 오프셋 값 이상 검출
008-88X	AxisX + Soft Limit	第 X 関節軸 +ソフトウェアリミット	第 X 轴 +软件限制	제 X 축 ＋소프트 리미트
008-89X	AxisX - Soft Limit	第 X 関節軸 -ソフトウェアリミット	第 X 轴 -软件限制	제 X 축 －소프트 리미트
008-90X	AxisX Position Backup Error	第 X 関節軸 位置バックアップ異常	第 X 轴 位置备份错误	제 X 축 위치 백업 이상
004-033	Interfere Error	干渉異常	干扰区域错误	간섭 영역 에러
004-034	Movement Limitation Area	動作制限エリア	分配位置 关节动作限制区域错误	동작 제한 구역
004-035	Outside Limitation Area	制限領域外	限制区外	제한 구역 밖
004-036	No Entry Area	立入禁止エリア	禁止进入区	출입 금지 구역
004-053	VSCALBIN parameter error	VSCALBIN パラメータ異常	VSCALBIN 命令 自变量错误	VSCALBIN 파라미터 이상
004-065	IP1 status alarm	IP1 ステータス異常	IP1 状态报警	IP1 상태 경고
004-066	IP2 status alarm	IP2 ステータス異常	IP2 状态报警	IP2 상태 경고
004-067	IP3 status alarm	IP3 ステータス異常	IP3 状态报警	IP3 상태 경고
004-068	IP4 status alarm	IP4 ステータス異常	IP4 状态报警	IP4 상태 경고
004-069	IP5 status alarm	IP5 ステータス異常	IP5 状态报警	IP5 상태 경고
004-070	IP6 status alarm	IP6 ステータス異常	IP6 状态报警	IP6 상태 경고
004-071	IP7 status alarm	IP7 ステータス異常	IP7 状态报警	IP7 상태 경고
004-072	IP8 status alarm	IP8 ステータス異常	IP8 状态报警	IP8 상태 경고
004-077	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
004-078	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
004-079	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
004-080	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
004-129	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-130	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-131	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-132	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-133	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-134	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-135	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-136	User Alarm	ユーザアラーム	用户报警	사용자 경고
004-42X	AxisX + Soft Limit	第 X 関節軸 +ソフトウェアリミット	第 X 轴 +软件限制	제 X 축 ＋소프트 리미트
004-43X	AxisX - Soft Limit	第 X 関節軸 -ソフトウェアリミット	第 X 轴 -软件限制	제 X 축 －소프트 리미트
002-017	Interpreter error	インタプリタ異常	解释器异常	인터프리터 이상
002-018	SVM Error	SVM 異常	SVM 异常	SVM 이상
002-019	Direct program error	ダイレクト実行失敗	不可直接执行程序	다이렉트 프로그램 실행 불가
002-020	Unable to Start	スタート不可	不可启动选择程序	셀렉트 프로그램 기동 불가
002-021	Unable to Continue	継続不可	不可继续执行选择程序	셀렉트 프로그램 계속 불가
002-022	Program Counter error	プログラムカウンタ異常	PC（行编号）异常	PC（행번호） 이상
002-023	Operate Code error	操作コード異常	操作代码异常	운영자 코드 이상
002-024	Operand error	オペランド異常	操作数异常	오퍼랜드 이상
002-025	Invalid command	コマンド異常	执行无效命令	무효 명령 실행
002-034	Ethernet connection error	Ethernet 接続異常	以太网连接错误	이더넷 접속 에러
002-035	Config data error	姿勢データ異常	姿势数据异常	자세 데이터 이상
002-036	Vector data error	ベクトルデータ異常	向量数据异常	벡터 데이터 이상
002-037	Strings too long	文字列長異常	字符串长度异常	문자열 길이 이상
002-039	PASS command error	PASS コマンド異常	CONV 参数异常	CONV 파라미터 이상
002-040	Invalid SMOOTH command	SMOOTH コマンド異常	CONV 参数异常	CONV 파라미터 이상
002-041	Invalid Variable	変数異常	无效变量区域	무효 변수 영역

Number	English	Japanese	Chinese	Korean
002-042	Data Type error	データ型異常	类型异常	타입 이상
002-043	Invalid ON command	ON コマンド異常	无效 ON 条件指定	무효 ON 조건 지정
002-045	Write to Const	定数領域書き込み	常数区域写入操作	정수 영역 기록 조작
002-046	Invalid Channel	チャンネル異常	无效通信通道	무효 통신 채널
002-047	Task Limits Over	タスク制限超過	超出指定任务数	지정 태스크수 초과
002-049	Symbol Overflow	シンボルオーバーフロー	符号空间溢出	심볼 영역 오버플로
002-050	Stack Overflow	スタックメモリオーバーフロー	堆栈空间溢出	스택 영역 오버플로
002-051	ON-DO Overflow	ON-DO オーバーフロー	ON-DO 溢出	ON 조건 동시 등록 초과
002-054	Text Overflow	テキストオーバーフロー	文本溢出	텍스트 오버플로
002-065	Motion command error	軌道生成演算異常	轨道生成运算错误	궤도 생성 연산 에러
002-066	Coordinate Conversion error	座標変換異常	不可转换目标位置坐标	목표 위치 좌표 변환 불가
002-067	MOVEC error	MOVEC 異常	圆弧轨道生成异常	원호 궤도 생성 이상
002-068	PAYLOAD error	PAYLOAD 異常	有效载荷参数异常	페이로드 파라미터 이상
002-069	CONFIG error	CONFIG 異常	不当姿势	부적합 자세
002-071	MOVE CV Calculation Error	MOVECV 計算エラー	MOVECV 计算错误	MOVECV 계산 오류
002-079	Point Out of limit	指令位置範囲外	偏离指定目标位置动作范围	지정 목표 위치 동작범위 외
002-081	Destination Interfere	指令位置干渉	指定位置干扰区域内	지정 위치 간섭 영역 내
002-082	Outside Limitation Area	制限領域外	定位限制区域	위치 결정 제한 영역
002-083	Movement Limitation Area	動作制限エリア	动作目标位置 关节动作限制区域错误	동작 제한 구역
002-084	Outside Limitation Area	制限領域外	限制区外	제한 구역 밖
002-085	No Entry Area	立入禁止エリア	禁止进入区	출입 금지 구역
002-089	MOVEJ Parameter error	MOVEJ パラメータ異常	MOVEJ 参数异常	MOVEJ 파라미터 이상
002-097	C Axis MOVE error	C 軸移動量超過	C 轴移动错误	C 축 이동량 초과
002-098	T Axis Impossible	T 軸不可	不可移动 T 轴	T 축 불가
002-129	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-130	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-131	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-132	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-133	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-134	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-135	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-136	User Alarm	ユーザアラーム	用户报警	사용자 경고
002-145	IP1 status alarm	IP1 ステータス異常	IP1 状态报警	IP1 상태 경고
002-146	IP2 status alarm	IP2 ステータス異常	IP2 状态报警	IP2 상태 경고
002-147	IP3 status alarm	IP3 ステータス異常	IP3 状态报警	IP3 상태 경고
002-148	IP4 status alarm	IP4 ステータス異常	IP4 状态报警	IP4 상태 경고
002-149	IP5 status alarm	IP5 ステータス異常	IP5 状态报警	IP5 상태 경고
002-150	IP6 status alarm	IP6 ステータス異常	IP6 状态报警	IP6 상태 경고
002-151	IP7 status alarm	IP7 ステータス異常	IP7 状态报警	IP7 상태 경고
002-152	IP8 status alarm	IP8 ステータス異常	IP8 状态报警	IP8 상태 경고
002-40X	AxisX Out of Limit	第 X 関節軸 動作制限範囲外	第 X 轴 动作限制范围外	제 X 축 목표 위치 동작범위 외
002-41X	AxisX Motor overload(15m)	第 X 関節軸 モータ過負荷(15 分)	第 X 轴 马达过载（15 分钟）	제 X 축 모터 오버로드(15 분)
002-42X	AxisX Motor overload(1m)	第 X 関節軸 モータ過負荷(1 分)	第 X 轴 马达过载（1 分钟）	제 X 축 모터 오버로드(1 분)
002-47X	AxisX Motor overload(2h)	第 X 関節軸 モータ過負荷(2 時間)	第 X 轴 马达过载（2 小时）	제 X 축 모터 과부하(2 시간)
002-56X	AxisX Overheat	第 X 関節軸 モータ温度超過	第 X 轴 马达温度超出	제 X 축 모터 온도 초과
001-003	M-TO Config Error	M-TO CONFIG 異常	示教点移动时配置错误	M-TO CONFIG 이상
001-009	Unable to RCYCLE	RCYCLE 不可	不可进行循环复位	사이클 리셋이 불가능
001-015	Parameter error	パラメータ異常	参数异常	파라미터 이상
001-017	PASS command error	PASS コマンド異常	通过命令无效	패스 커맨드의 무효
001-018	Invalid SMOOTH command	SMOOTH コマンド異常	平滑命令无效	스무드 커맨드의 무효
001-025	Over Deceleration	減速超過	超出减速加速度	감속 가속도 오버
001-026	SMOOTH impossible	SMOOTH 不可	不可执行 SMOOTH 命令	SMOOTH 커맨드 불가능
001-028	Already Enable SW ON	イネーブルスイッチオン	启用开关 ON 状态	이네이블 스위치 ON 상태
001-037	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-038	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고

Number	English	Japanese	Chinese	Korean
001-039	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-040	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-041	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-042	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-043	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-044	PLC user alarm	PLC ユーザアラーム	PLC 用户报警	PLC 사용자 경고
001-082	Option1 I/O Module Detect	Option1 I/O モジュール検出	Option1 I/O 基板检测	Option1 I/O 기판 검출
001-083	Option3 Conveyor Cable Detect	Option3 コンベア同期用ケーブル検出		
001-084	Option4 AD Module Detect	Option4 AD モジュール検出	Option4 AD 基板检测	Option4 AD 기판 검출
001-085	Hand1 Module Detect	ハンド 1 モジュール検出	Hand1 基板检测	Hand1 기판 검출
001-097	Initial IP address mode	初期ＩＰアドレスモード	初始 IP 地址模式	초기 IP 주소 모드
001-099	FRAM0 erase mode	FRAM0 消去モード		
001-101	Parameter file deletion mode	パラメータファイル削除モード	参数文件删除模式	파라미터 파일 삭제 모드
001-108	Memory erase mode	メモリ消去モード	存储器擦除模式	메모리 삭제 모드
001-129	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-130	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-131	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-132	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-133	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-134	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-135	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-136	User Alarm	ユーザアラーム	用户报警	사용자 경고
001-145	MAIN Module Battery alarm	メインモジュールバッテリ異常	主电池报警	메인 배터리 경고
001-147	COM Buffer Over	COM バッファ超過	通信缓冲区溢出	통신 버퍼 오버플로
001-149	Extern SELECT error	Extern SELECT 異常	外部文件选择错误	외부 파일 선택 에러
001-150	Data block duplication error	データ部重複異常	数据部分重复异常	데이터부 중복 이상
001-154	Data block format error	データ部フォーマット異常	数据块格式错误	데이터 블록 포맷 에러
001-158	Short of working area	編集作業領域不足	操作空间不足	동작 영역 부족
001-160	Compile error	コンパイル異常	编译错误	컴파일 에러
001-164	PLC Backup data error	PLC バックアップデータ異常	PLC 备份数据错误	PLC 백업 데이터 에러
001-166	PLC Remote unit error	PLC リモート装置異常	PLC 远程单元错误	PLC 리모트 유닛 에러
001-169	PLC undefine label	PLC 未定義ラベル	PLC 未定义标签	PLC 미정의 라벨
001-170	PLC invalid command	PLC コマンド異常	PLC 无效命令	PLC 무효 커맨드
001-173	PLC Overlap label	PLC ラベル重複	PLC 标签重复	PLC 라벨 중복
001-194	Can not Use SD Card	SD カード使用不可	不可使用 SD 卡	SD 카드 사용 불가
001-201	IP1 status alarm	IP1 ステータス異常	IP1 状态报警	IP1 상태 경고
001-202	IP2 status alarm	IP2 ステータス異常	IP2 状态报警	IP2 상태 경고
001-203	IP3 status alarm	IP3 ステータス異常	IP3 状态报警	IP3 상태 경고
001-204	IP4 status alarm	IP4 ステータス異常	IP4 状态报警	IP4 상태 경고
001-205	IP5 status alarm	IP5 ステータス異常	IP5 状态报警	IP5 상태 경고
001-206	IP6 status alarm	IP6 ステータス異常	IP6 状态报警	IP6 상태 경고
001-207	IP7 status alarm	IP7 ステータス異常	IP7 状态报警	IP7 상태 경고
001-208	IP8 status alarm	IP8 ステータス異常	IP8 状态报警	IP8 상태 경고
001-209	Robot S/N Mismatch	ロボットシリアル番号不一致	机器人序列号不匹配	로봇 일련 번호 불일치
001-40X	AxisX Encoder Battery low	第 X 関節軸 エンコーダバッテリ低下	第 X 轴 位置检测器电池报警	제 X 축 위치 검출기 배터리 경고
001-42X	AxisX Failed to Read from Encoder EEPROM	第 X 関節軸 エンコーダ EEPROM 読込失敗	第 X 轴 编码器 EEPROM 读取失败	제 X 축 인코더 EEPROM 읽기 실패
001-43X	AxisX Failed to Write to Encoder EEPROM	第 X 関節軸 エンコーダ EEPROM 書込失敗	第 X 轴 编码器 EEPROM 写入失败	제 X 축 인코더 EEPROM 쓰기 실패